Where Deep Learning Losses Come From

Likelihood, cross-entropy, and regularization from probability

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ES 667 · Deep Learning · IIT Gandhinagar

Three familiar losses, one probabilistic origin

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Objective. Derive each term from an explicit likelihood or prior.

The map for today

V VISUAL

Two probabilistic assumptions determine the two halves of the objective.

We will define the predictive distribution and likelihood; use i.i.d. to move from one example to a dataset;
then introduce log-likelihood, NLL, the prior $p(\boldsymbol{\theta})$, and MAP.

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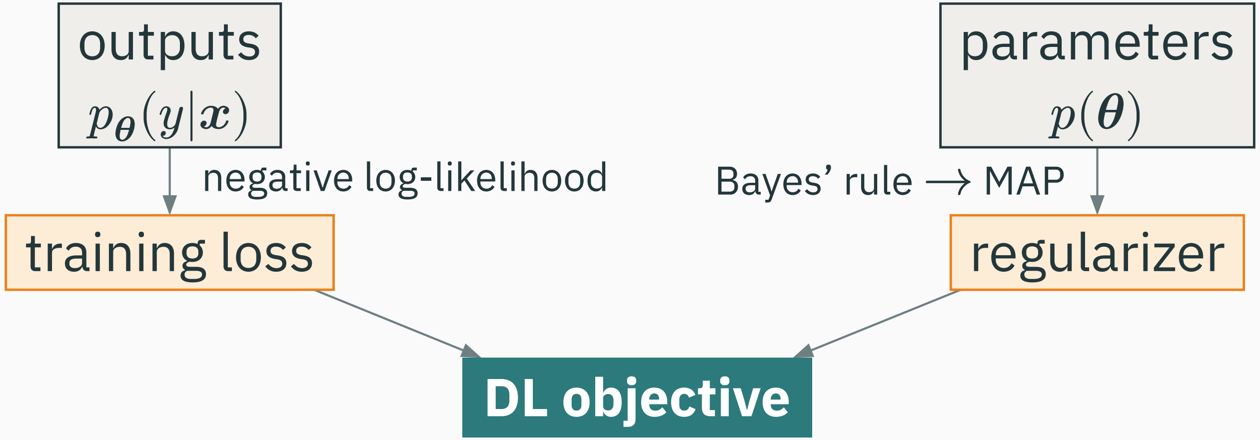


Probability primer

A point prediction summarizes a distribution

V VISUAL

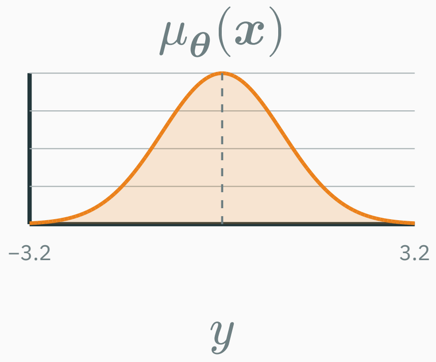
REGRESSION · $Y \in \mathbb{R}$

Point prediction: mean of $Y \mid \boldsymbol{X} = \boldsymbol{x}$

$$\hat{y}(\boldsymbol{x}) = \mathbb{E}_{\boldsymbol{\theta}}[Y \mid \boldsymbol{X} = \boldsymbol{x}] = \mu_{\boldsymbol{\theta}}(\boldsymbol{x})$$

Full predictive distribution

$$Y \mid \boldsymbol{X} = \boldsymbol{x} \sim \mathcal{N}\big(\mu_{\boldsymbol{\theta}}(\boldsymbol{x}), \sigma^2_{\boldsymbol{\theta}}(\boldsymbol{x})\big)$$



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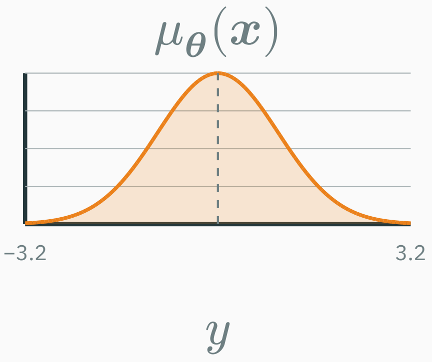

BINARY CLASSIFICATION · $Y \in \{0, 1\}$

Point prediction: mode of $Y \mid \boldsymbol{X} = \boldsymbol{x}$

$$\hat{y}(\boldsymbol{x}) = \mathbf{1}[p_{\boldsymbol{\theta}}(\boldsymbol{x}) \geq 0.5]$$

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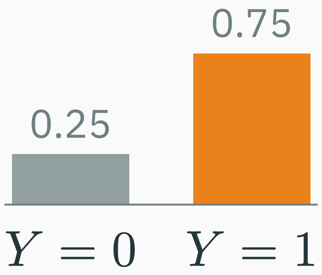
$$Y \mid \boldsymbol{X} = \boldsymbol{x} \\ \sim \mathrm{Bernoulli}(p_{\boldsymbol{\theta}}(\boldsymbol{x}))$$

example: $p_{\boldsymbol{\theta}}(\boldsymbol{x}) = 0.75$



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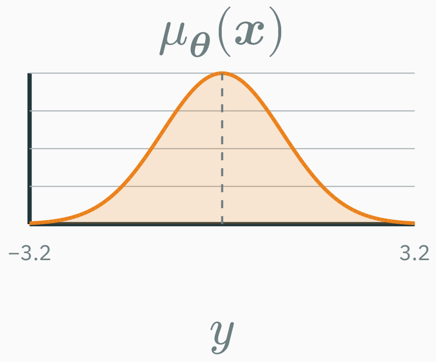

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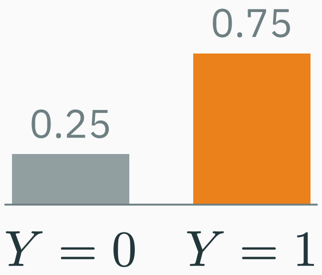


Neural network: $x \rightarrow$ distribution parameters. Prediction: $\hat{y} =$ mean or mode.

Discrete distributions assign probability to outcomes

V VISUAL

The **support** $\mathcal{S}_Y$ is the set of values that Y can take.

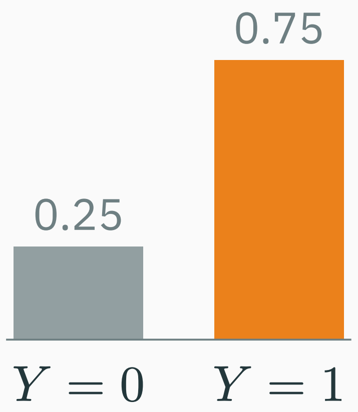
Probability mass function (PMF)

$$p_{Y(y)} = P(Y = y), \quad y \in \mathcal{S}_Y$$

$$p_{Y(y)} \geq 0, \quad \sum_{y \in \mathcal{S}_Y} p_{Y(y)} = 1$$

Examples. Bernoulli: $\{0, 1\}$ · categorical: $\{1, \ldots, K\}$ · Poisson: $\{0, 1, 2, \ldots\}$

Bernoulli(0.75) PMF



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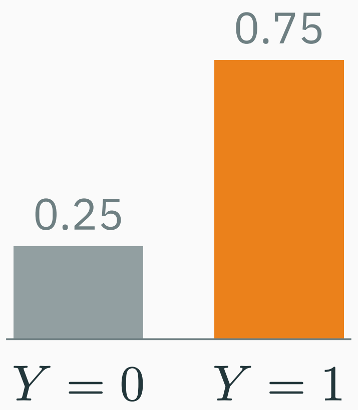

each bar is an outcome probability

Exact values can have nonzero probability: here $P(Y = 1) = 0.75$.

Continuous distributions assign density across a support

V VISUAL

The **support** may be bounded or unbounded:

$$\mathcal{S}_Y = [a, b]$$

for a Uniform variable; $\mathcal{S}_Y = \mathbb{R}$ for a Gaussian.

Probability density function (PDF)

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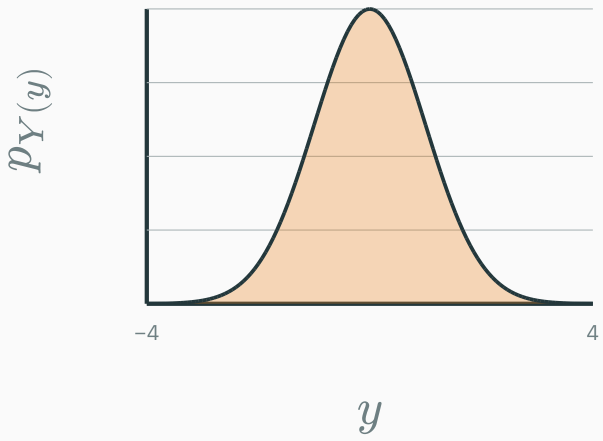

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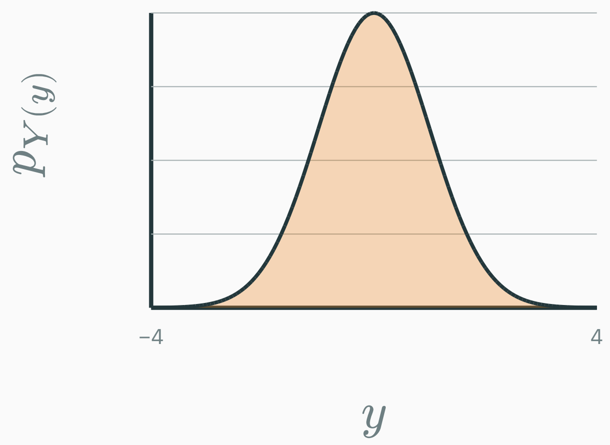

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For continuous Y, probability comes from area under the density.

Density is not probability

V VISUAL

For a continuous Y, probability comes from **area**:

$$P(a \leq Y \leq b) = \int_a^b p_{Y(y)} \, \mathrm{d}y$$

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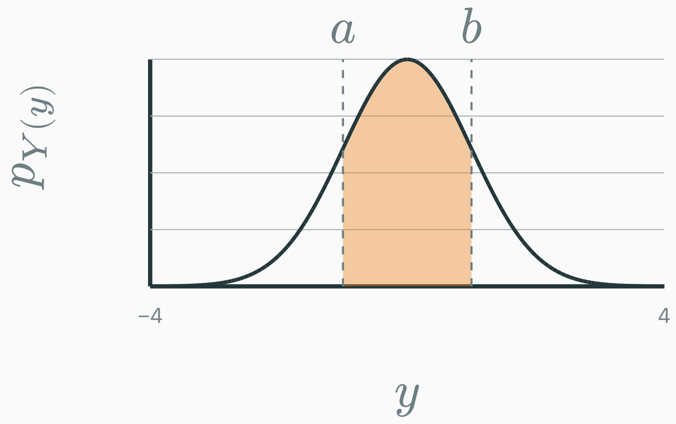
$p_{Y(y)} \neq P(Y = y)$ — a density can **exceed 1**; only *integrals* are probabilities.



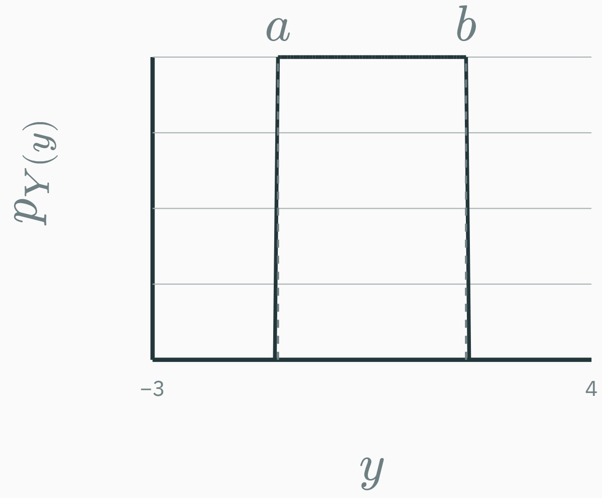
the shaded area is $P(a \leq Y \leq b)$

Uniform distribution: support matters

V VISUAL

$$Y \sim \mathcal{U}(a,b) \quad p_{Y(y)} = \begin{cases} 1/(b-a) & a \le y \le b \\ 0 & \text{otherwise} \end{cases}$$

> The support matters: outside $[a, b]$, the density is exactly 0.



Gaussian distribution

V VISUAL

$$Y \sim \mathcal{N}(\mu, \sigma^2) \qquad p(y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right)$$

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Only two knobs: **location** μ and **scale** σ.

From probability to likelihood

Read the model notation one symbol at a time

Q QUESTION

$$p_{\boldsymbol{\theta}}(y \mid \boldsymbol{x})$$

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$\boldsymbol{X}$ / $\boldsymbol{x}$	$\boldsymbol{X}$ is the random input vector; $\boldsymbol{x}$ is one observed input
Y / y	Y is the random target; y is one possible or observed value
$\boldsymbol{\theta}$	the **parameter vector** — the model's learned weights and biases
$p_{\boldsymbol{\theta}}(y \mid \boldsymbol{x})$	the probability (discrete Y) or density (continuous Y) assigned to y, **given** $\boldsymbol{x}$

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Read aloud: "model θ's probability or density for y, given x."

Prediction fixes the model and varies the outcome

V VISUAL

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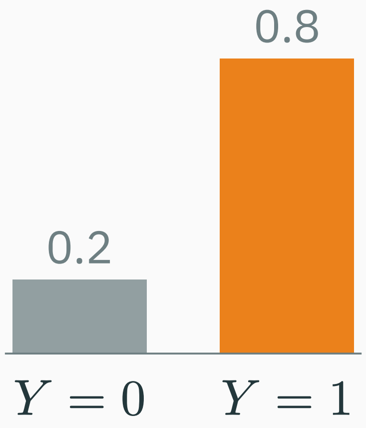
$$p_{\boldsymbol{\theta}_A}(Y = 0 \mid \boldsymbol{x}^\star) = 0.20$$

$$p_{\boldsymbol{\theta}_A}(Y = 1 \mid \boldsymbol{x}^\star) = 0.80$$

$$0.20 + 0.80 = 1$$

We vary the possible value of Y while keeping $\boldsymbol{\theta}_A$ and $\boldsymbol{x}^\star$ fixed.

predictive distribution



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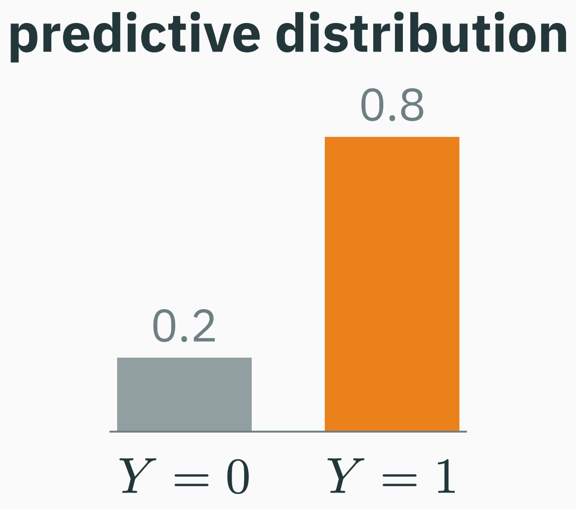

$$\boxed{\hat{y}(\boldsymbol{x}^\star) = \arg\max_y p_{\boldsymbol{\theta}_A}(y \mid \boldsymbol{x}^\star) = 1}$$

One observed label gives one likelihood score

D DERIVATION

Now observe $y^\star = 1$ for the input $\boldsymbol{x}^\star$, so $\mathcal{D} = \{(\boldsymbol{x}^\star, 1)\}$.

$$L(\boldsymbol{\theta}; \mathcal{D}) = p_{\boldsymbol{\theta}}(y^\star = 1 \mid \boldsymbol{x}^\star)$$

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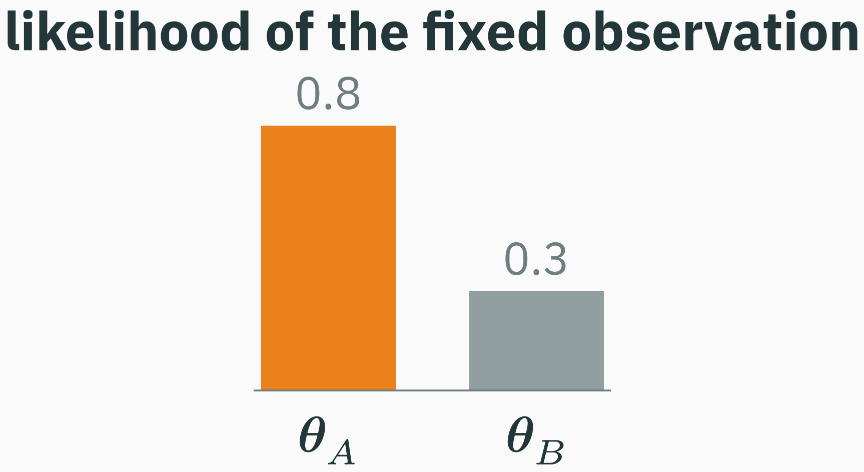
> $\boldsymbol{\theta}_A$ makes the observed label $y^\star = 1$ less surprising.



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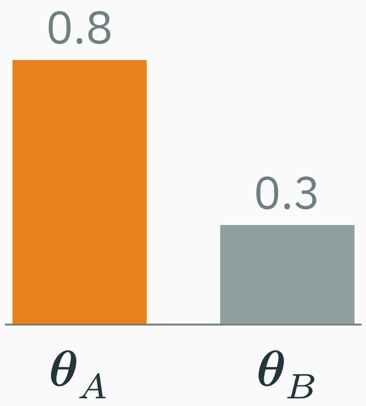
likelihood of the fixed observation



Likelihood is a **score as a function of θ**; the scores need not sum to 1 over parameter settings.

One continuous observation gives a likelihood too

D DERIVATION

Suppose $Y \mid \mu \sim \mathcal{N}(\mu, 1)$ and we observe the fixed value $y^\star = 1$.

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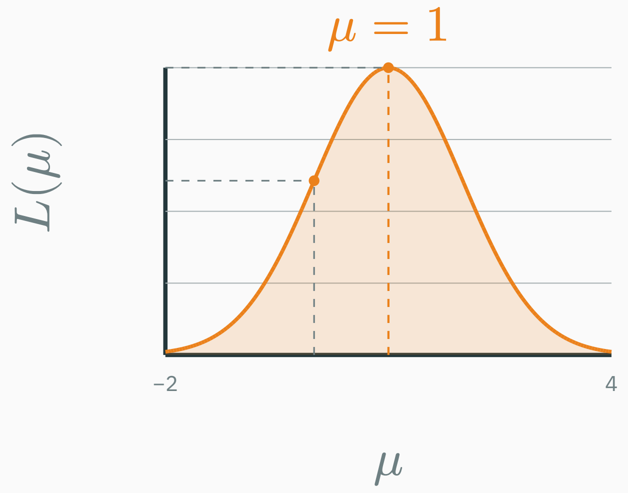
Candidate $\mu = 0$: $L(0) \approx 0.242$

$\mu = 1$ centres the Gaussian at $y^\star = 1$ and gives the largest **density**.



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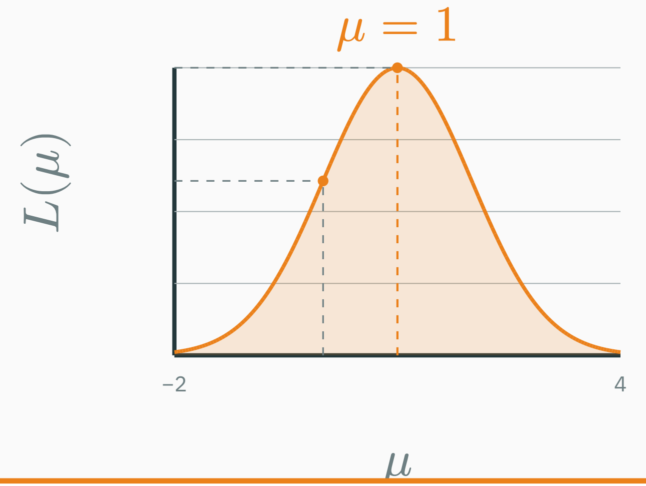

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Discrete or continuous: fix what was observed, then ask which parameter values explain it better.

A dataset needs a joint likelihood

Q QUESTION

For two observations, the likelihood starts as a **joint** probability or density:

$$L(\boldsymbol{\theta}; \mathcal{D}) = p_{\boldsymbol{\theta}}(y_1, y_2 \mid \boldsymbol{x}_1, \boldsymbol{x}_2).$$

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The product is a consequence of **conditional independence**; it is not automatic.

Independent means “no extra information”

D DERIVATION

After fixing θ, observing one outcome does not change the distribution of another:

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Coin-flip example. If $P(H \mid \theta) = \theta$ and the flips are independent, then $P(H, H \mid \theta) = \theta^2$.

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Independence is what licenses multiplication.

Identically distributed means “the same sampling rule”

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Coin flips

$Y_i \mid \theta \sim \mathrm{Bernoulli}(\theta)$ for every i.
Each flip uses the same bias θ.

Supervised learning

The pairs $(\boldsymbol{X}_i, Y_i)$ come from the same population, and one shared $\boldsymbol{\theta}$ defines $p_{\boldsymbol{\theta}}(y_i \mid \boldsymbol{x}_i)$.

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i.i.d. = independent observations + one common data-generating rule.

Independence and identical distribution can fail separately

D DERIVATION

Not independent: daily weather

Tomorrow’s weather depends on today’s. A first-order Markov model writes

$$p(y_{1:T} \mid \theta) = p(y_1 \mid \theta) \prod_{t=2}^{T} p(y_t \mid y_{t-1}, \theta).$$

The factors are **conditional on the previous day**, not independent marginals.

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Independent but not identical: two sensors

Two sensors measure the same temperature μ independently, but have different noise levels:

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i.i.d. is a useful special case—not a universal law.

Worked example: dependence changes the product

D DERIVATION

Let R_1 mean "rain today" and R_2 mean "rain tomorrow". Suppose

$$P(R_1) = 0.4, \quad P(R_2|R_1) = 0.8, \quad P(R_2|\neg R_1) = 0.2.$$

1. Use the conditional rule

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Both days are rainy with probability 0.32.

Worked example: dependence changes the product

D DERIVATION

Let R_1 mean "rain today" and R_2 mean "rain tomorrow". Suppose

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Dependence does not forbid multiplication: use a conditional probability.

Worked example: independent does not mean identical

D DERIVATION

Given smoke S, two alarms act independently, but their sensitivities differ:

$$P(A_1 = 1|S) = 0.9, \quad P(A_2 = 1|S) = 0.6.$$

Independence

Both alarms ring with probability

$$P(A_1 = 1, A_2 = 1|S)$$
$$= 0.9 \times 0.6 = 0.54.$$

The product is valid.

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Alarm 1 detects smoke 90% of the time;
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Independent says “multiply”; identical says “reuse the same sampling rule”.

IID turns the joint likelihood into per-example factors

D DERIVATION

For i.i.d. supervised examples, treat the observed inputs as fixed. Conditional independence gives

$$p(\mathcal{D} \mid \boldsymbol{\theta}) = \prod_{i=1}^{n} p_{\boldsymbol{\theta}}(y_i \mid \boldsymbol{x}_i).$$

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This is the bridge from one-example likelihood to a dataset loss.

Work two tiny Bernoulli datasets

D DERIVATION

Compare a fair coin $\theta = 0.5$ with a head-biased coin $\theta = 0.8$.

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Observed data	$L(0.5)$	$L(0.8)$	Which explains it better?
H, T	$0.5(0.5) = 0.25$	$0.8(0.2) = 0.16$	fair coin
H, H	$0.5^2 = 0.25$	$0.8^2 = 0.64$	head-biased coin

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Multiply one Bernoulli term per independent flip.

Two Gaussian measurements: multiply densities

D DERIVATION

Let $Y_i \mid \mu \sim \mathcal{N}(\mu, 1)$ independently, and observe $y_1 = 1$, $y_2 = 2$.

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Candidate $\mu = 1.5$

$p(y_1|\mu) \approx 0.352$

$p(y_2|\mu) \approx 0.352$

$L(1.5) \approx 0.352 \times 0.352 = 0.124$

Candidate $\mu = 0$

$p(y_1|\mu) \approx 0.242$

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$L(0) \approx 0.242 \times 0.054 = 0.013$

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For continuous data we multiply densities; $\mu = 1.5$ gives the larger likelihood.

Companion notebook: likelihood from one example to a dataset

I INTERACTIVE

INTERACTIVE

↗ github.com/nipunbatra/dl-teaching/blob/master/notebooks/L01/00_likelihood_iid_worked_examples.ipynb

Compute and plot single-observation likelihoods, multiply independent Bernoulli and Gaussian terms, compare candidate parameters, and verify that log-likelihood is a sum.

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In class: predict the winning parameter first; then run the cell and explain why the curve peaks there.

Coin flips: a likelihood you can compute

D DERIVATION

Observed: $H, H, T, T, T, H, H, T, T, T$

Hence $n_H = 4$ and $n_T = 6$.

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$$L(\theta) = \prod_i p(x_i \mid \theta)$$

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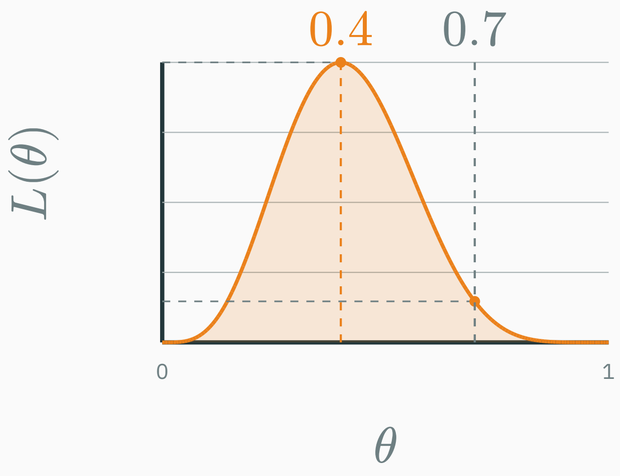
$$= \theta^4(1-\theta)^6$$



$$L(0.4) \approx 0.00119$$
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about $6.8 \times$ larger at $\theta = 0.4$

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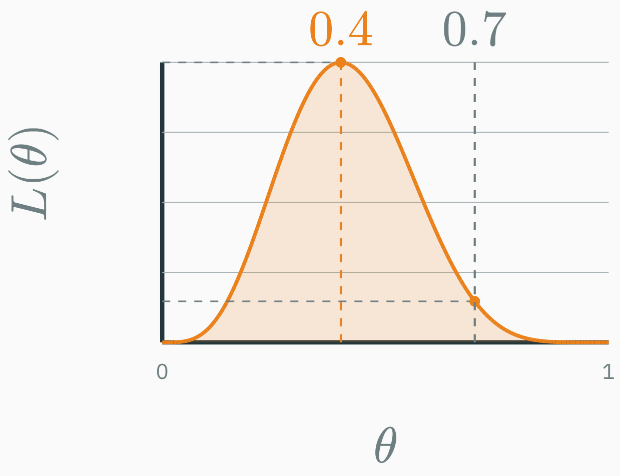

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The likelihood peaks at $\theta = 0.4$ — the observed head fraction $\frac{4}{10}$.

Coin flips: calculate the log-likelihood

D DERIVATION

The **log-likelihood** is the logarithm of the likelihood — not a different model:

$$\ell(\theta) := \log L(\theta) = \log\left[\theta^4(1-\theta)^6\right] = 4\log\theta + 6\log(1-\theta)$$

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Likelihoods below 1 have negative logs; among them, -6.73 is the larger value.

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The observed sequence has higher log-likelihood at $\theta = 0.4$.

Why logs? The maximizing parameter does not change

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$$a > b > 0 \implies \log a > \log b$$

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Both likelihood and log-likelihood rank $\theta = 0.4$ above $\theta = 0.7$.

Poisson example: logging preserves the peak

V VISUAL

Suppose $Y \sim \mathrm{Poisson}(\lambda)$ and the observed count is $y = 3$.

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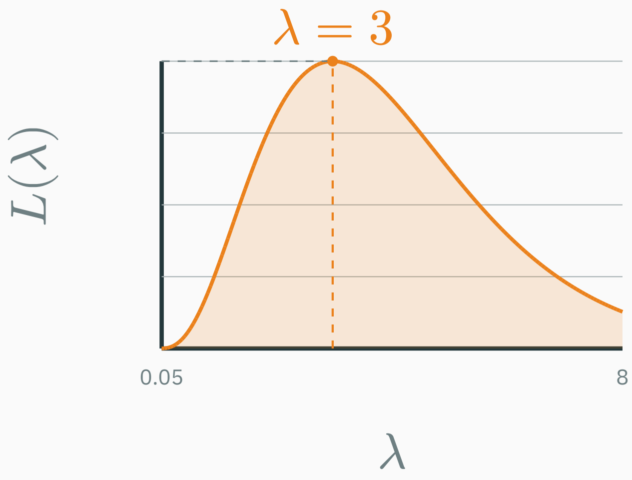
$$L(\lambda) = e^{-\lambda}\frac{\lambda^3}{3!}$$



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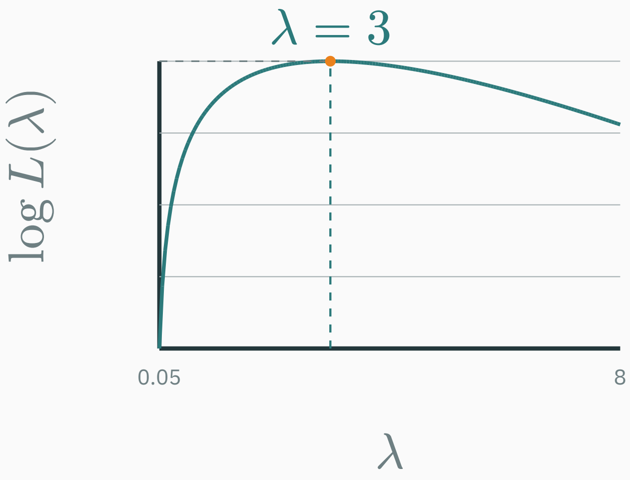

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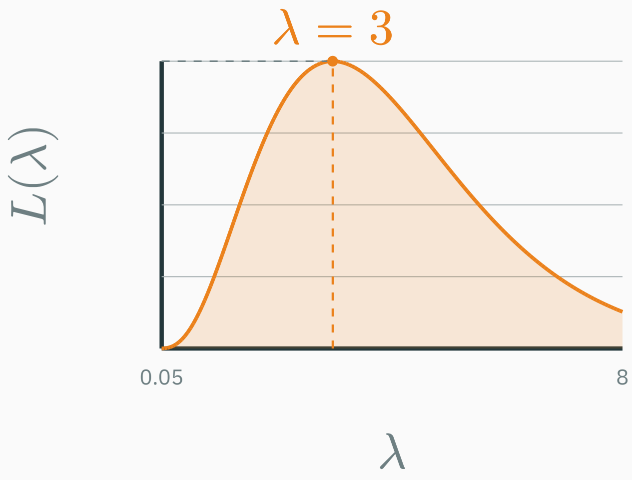

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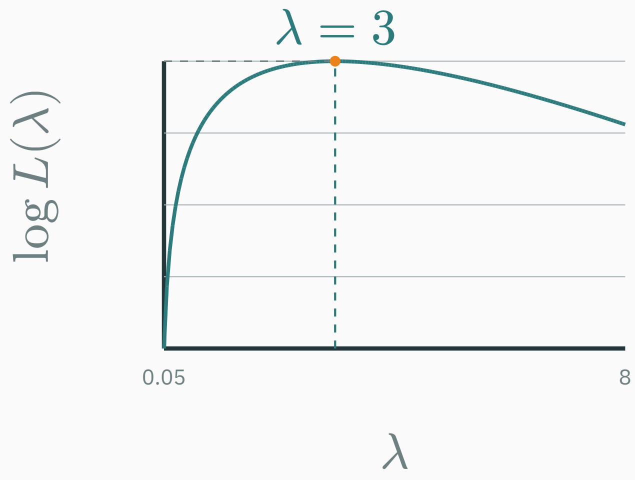

Both curves peak at $\hat{\lambda}_{\mathrm{MLE}} = 3$: the log changes the vertical scale, not the maximizing parameter.

Why logs? Products become stable sums

DERIVATION

For independent observations,

$$L(\boldsymbol{\theta}) = \prod_i p_{\boldsymbol{\theta}}(y_i \mid \boldsymbol{x}_i) \quad \Longrightarrow \quad \log L(\boldsymbol{\theta}) = \sum_i \log p_{\boldsymbol{\theta}}(y_i \mid \boldsymbol{x}_i)$$

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Worked numerical example. Suppose 200 observed outcomes each receive probability 0.01.

Direct product	$L = 0.01^{200} = 10^{-400} < 4.94 \times 10^{-324}$ → underflows to 0
Log space	$\log L = 200 \log(0.01) \approx -921.03$ → finite and representable

Float64 range: $\pm 1.80 \times 10^{308}$ · smallest positive normal 2.23×10^{-308} · smallest subnormal 4.94×10^{-324}.

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Logs make optimization easier and prevent numerical underflow.

Coin flips: solve for the stationary point

D DERIVATION

For $L(\theta) = \theta^{n_H}(1-\theta)^{n_T}$, the log-likelihood is

$$\log L(\theta) = n_H \log \theta + n_T \log(1-\theta)$$

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Set the derivative to zero:

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$\theta^\star = 0.4$ **is a candidate optimum. Next, check the curvature.**

The second derivative confirms a maximum

D DERIVATION

Differentiate the log-likelihood again:

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At the stationary point $\theta^\star = 0.4$,

$$-\frac{4}{(0.4)^2} - \frac{6}{(0.6)^2} = -25 - 16.67 \approx -41.67 < 0$$

Strict concavity $+$ stationary point $\rightarrow$ unique global maximum: $\hat{\theta}_{\text{MLE}} = \theta^\star = \frac{4}{10} = 0.4$**.**

Maximum likelihood estimation

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Maximum likelihood estimation

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Interpretation. Choose the parameter values that assign the **highest likelihood** to the observed dataset.

Maximizing likelihood = minimizing NLL

D DERIVATION

$$\arg\max_{\boldsymbol{\theta}} \prod_i p_{\boldsymbol{\theta}}(y_i \mid \boldsymbol{x}_i) = \arg\max_{\boldsymbol{\theta}} \sum_i \log p_{\boldsymbol{\theta}}(y_i \mid \boldsymbol{x}_i)$$

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Per example	$\ell_{i(\boldsymbol{\theta})} = -\log p_{\boldsymbol{\theta}}(y_i \mid \boldsymbol{x}_i)$
Training set	$\hat{R}(\boldsymbol{\theta}) = \frac{1}{n}\sum_i \ell_{i(\boldsymbol{\theta})} = -\frac{1}{n}\log p(\mathcal{D} \mid \boldsymbol{\theta})$

The final equality uses conditional independence: $p(\mathcal{D}|\boldsymbol{\theta}) = \prod_i p_{\boldsymbol{\theta}}(y_i|\boldsymbol{x}_i)$.

Empirical risk = average training loss

D DERIVATION

In ML, **risk** means expected loss. Because the data-generating distribution is unknown, we estimate it from the observed training set.

Empirical risk = average loss on $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^{n}$.

Per example	$\ell_{i(\boldsymbol{\theta})} = -\log p_{\boldsymbol{\theta}}(y_i \mid \boldsymbol{x}_i)$
Training set	$\hat{R}(\boldsymbol{\theta}) = \frac{1}{n}\sum_i \ell_{i(\boldsymbol{\theta})} = -\frac{1}{n}\log p(\mathcal{D} \mid \boldsymbol{\theta})$

The final equality uses conditional independence: $p(\mathcal{D}|\boldsymbol{\theta}) = \prod_i p_{\boldsymbol{\theta}}(y_i|\boldsymbol{x}_i)$.

Minimizing empirical NLL is maximum likelihood; $\frac{1}{n}$ only rescales the objective.

Worked example: support becomes a hard loss

D DERIVATION

Recall the Uniform observation model:

$$p_{\boldsymbol{\theta}}(y_i \mid \boldsymbol{x}_i) = \begin{cases} 1/(b-a) & a \le y_i \le b \\ 0 & \text{otherwise} \end{cases}$$

Worked example: support becomes a hard loss

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Zero likelihood becomes an infinite training penalty.

Checkpoint: uniform support

Q QUESTION

For a Uniform(a, b) observation model, what is the NLL if the target lies outside $[a, b]$?

A. 0

B. A fixed finite penalty

C. $-\log 0 = +\infty$

D. It depends only on σ

Answer: uniform support

A ANSWER

Correct: C — $-\log 0 = +\infty$

Why: The model assigns zero density outside its support, so the observation is impossible under the assumption.

Regression: why MSE?

Regression as signal plus noise

$$y_i = f_{\boldsymbol{\theta}}(\boldsymbol{x}_i) + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

Regression as signal plus noise

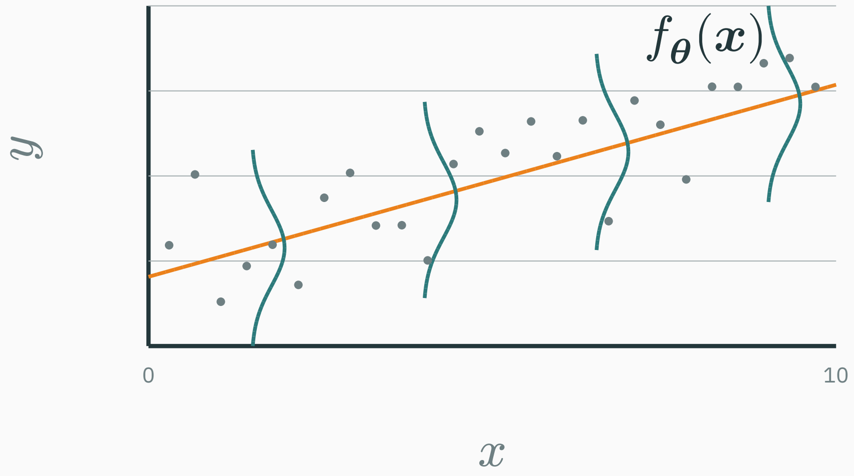
$$y_i = f_{\boldsymbol{\theta}}(\boldsymbol{x}_i) + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

Therefore the conditional output distribution is

$$Y_i \mid \boldsymbol{X}_i = \boldsymbol{x}_i \sim \mathcal{N}\big(f_{\boldsymbol{\theta}}(\boldsymbol{x}_i), \sigma^2\big)$$

What the assumption means visually

V VISUAL



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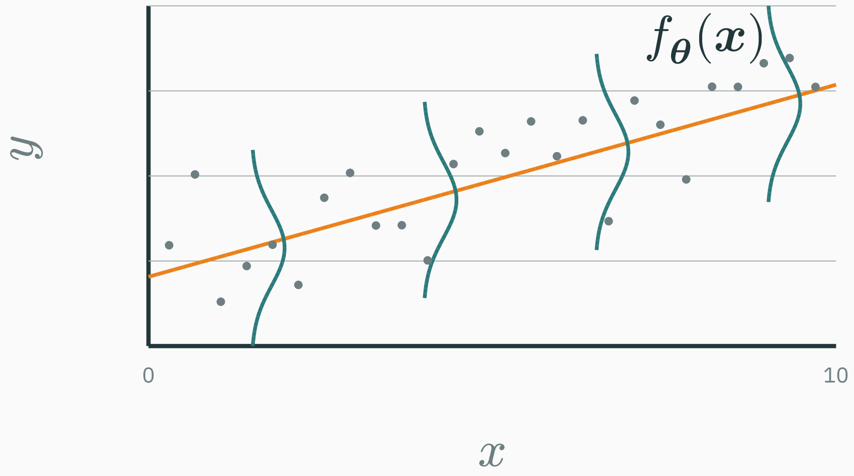

Gaussian noise lives in the **output direction**, conditional on x — a vertical bell at every input.

Gaussian likelihood

D DERIVATION

$$p_{\boldsymbol{\theta}}(y_i \mid \boldsymbol{x}_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - f_{\boldsymbol{\theta}}(\boldsymbol{x}_i))^2}{2\sigma^2}\right)$$

Gaussian likelihood

D DERIVATION

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Negative log of a single example:

$$-\log p_{\boldsymbol{\theta}}(y_i \mid \boldsymbol{x}_i) = \frac{\left(y_i - f_{\boldsymbol{\theta}}(\boldsymbol{x}_i)\right)^2}{2\sigma^2} + C$$

MSE is Gaussian MLE

Sum over the dataset:

$$-\log p(\mathcal{D} \mid \boldsymbol{\theta}) = \frac{1}{2\sigma^2} \sum_i \left(y_i - f_{\boldsymbol{\theta}}(\boldsymbol{x}_i)\right)^2 + C$$

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With σ fixed, the minimizer is

$$\hat{\boldsymbol{\theta}}_{\text{MLE}} = \arg\min_{\boldsymbol{\theta}} \sum_i \left(y_i - f_{\boldsymbol{\theta}}(\boldsymbol{x}_i)\right)^2$$

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MSE = Gaussian NLL, up to constants.

Linear regression makes f_θ explicit

D DERIVATION

For the one-dimensional picture,

$$\hat{y}_i = \theta_0 + \theta_1 x_i$$

Linear regression makes f_θ explicit

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For the one-dimensional picture,

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For d features, prepend a constant 1 so the intercept is part of the same parameter vector:

$$\tilde{\boldsymbol{x}}_i = (1, \boldsymbol{x}_i), \quad \boldsymbol{\theta} = (\theta_0, \theta_1, ..., \theta_d)$$

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Then the same notation used throughout the lecture becomes

$$\hat{y}_i = f_{\boldsymbol{\theta}}(\boldsymbol{x}_i) = \boldsymbol{\theta}^\top \tilde{\boldsymbol{x}}_i$$

θ_0 is the intercept; $\theta_1, \ldots, \theta_d$ are the slopes.

Gaussian MLE becomes ordinary least squares

D DERIVATION

Substitute the linear mean into the observation model:

$$Y_i \mid \boldsymbol{X}_i = \boldsymbol{x}_i \sim \mathcal{N}(\boldsymbol{\theta}^\top \tilde{\boldsymbol{x}}_i, \sigma^2)$$

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$$r_i = y_i - \hat{y}_i = y_i - \boldsymbol{\theta}^\top \tilde{\boldsymbol{x}}_i$$

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With σ fixed, maximum likelihood therefore gives

$$\hat{\boldsymbol{\theta}}_{\text{MLE}} = \arg\min_{\boldsymbol{\theta}} \sum_i \underbrace{r_i^2}_{\text{squared vertical error}}$$

Ordinary least squares is Gaussian maximum likelihood for a linear mean.

The normal equations solve linear regression

D DERIVATION

Stack the augmented inputs into $\tilde{\boldsymbol{X}} = [\mathbf{1} \quad \boldsymbol{X}]$. Then

$$\hat{\boldsymbol{y}} = \tilde{\boldsymbol{X}}\boldsymbol{\theta}, \quad J(\boldsymbol{\theta}) = \|\boldsymbol{y} - \tilde{\boldsymbol{X}}\boldsymbol{\theta}\|_2^2$$

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This gives the **normal equations**:

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If $\tilde{\boldsymbol{X}}^\top \tilde{\boldsymbol{X}}$ is invertible,

$$\hat{\boldsymbol{\theta}} = \left(\tilde{\boldsymbol{X}}^\top \tilde{\boldsymbol{X}}\right)^{-1} \tilde{\boldsymbol{X}}^\top \boldsymbol{y}$$

Linear regression has a closed form; neural networks keep the MSE objective but use iterative optimization.

Why earlier ML courses could ignore σ^2

Q QUESTION

σ^2 is the observation-noise variance: the expected squared spread of targets around the mean prediction.

For n Gaussian observations, write $r_i = y_i - \mu_i$. The dataset NLL is

$$\mathcal{L}(\boldsymbol{\theta}; \sigma^2) = \frac{1}{2\sigma^2} \sum_i r_i^2 + \frac{n}{2} \log \sigma^2 + C.$$

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For point predictions, fixed-variance Gaussian MLE gives the same fitted model as MSE.

How is σ^2 chosen in practice?

D DERIVATION

One global noise level.

Fit the mean model, then estimate the residual spread:

$$\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{\mu}_i)^2.$$

Use this when a residual plot has roughly the same vertical width across inputs.

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Input-dependent noise.

Let the network predict both

$$(\mu_i, s_i) = f_{\boldsymbol{\theta}}(\boldsymbol{x}_i),$$

$$s_i = \log \sigma_i^2, \quad \sigma_i^2 = e^{s_i} > 0.$$

Use this when residual spread changes with the input—for example, a funnel shape.

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Fit both cases with Gaussian NLL; validate whether uncertainty intervals attain their promised coverage.

Homoscedastic vs input-dependent uncertainty

V VISUAL

HOMOSCEDASTIC

$$\boldsymbol{x}_i \longrightarrow f_{\boldsymbol{\theta}} \longrightarrow \mu_i \longrightarrow \mathcal{N}(\mu_i, \sigma^2)$$

Network output: μ_i

Shared variance: σ^2

same uncertainty at every $\boldsymbol{x}_i$

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HETEROSCEDASTIC

$$\begin{tikzcd}
 & & \mu_i \arrow[dr] & \\
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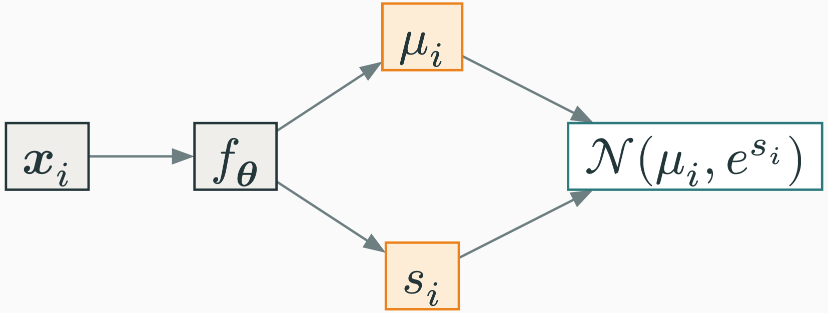

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uncertainty changes with $\boldsymbol{x}_i$

Predict log-variance so every real s_i maps to a positive variance e^{s_i}.

The noise model determines how errors are penalized

V VISUAL

Noise model	Negative log-likelihood
Gaussian	r^2
Laplace	$\lvert r \rvert$
Uniform	0 inside; $+\infty$ outside
Student-t	$\log\left(1 + \frac{r^2}{\nu}\right)$

$r = y - \hat{y}$ · swatches match plot

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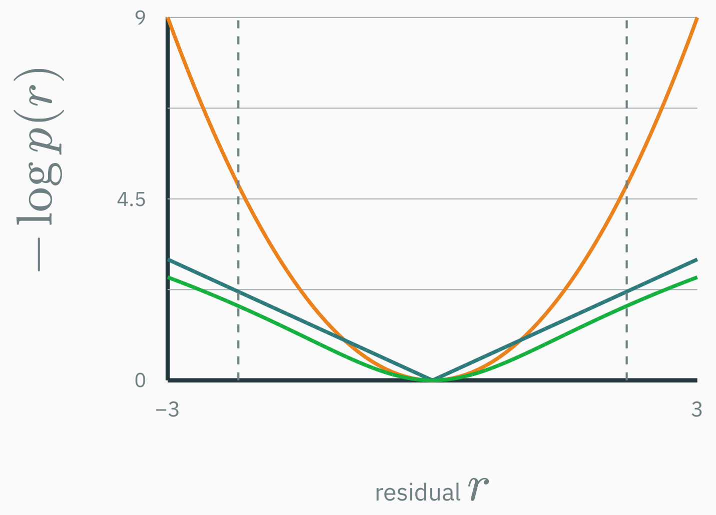

At $r = 0$, prediction equals target. Farther horizontally = larger error; higher vertically = larger NLL penalty.

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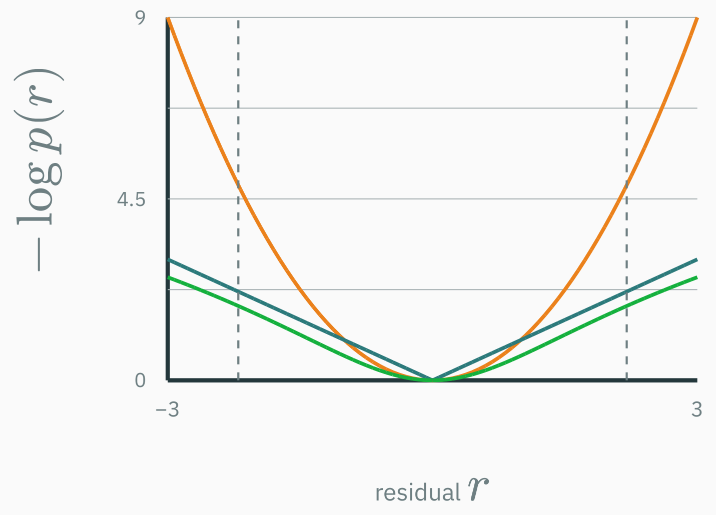

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Gaussian penalizes large errors most; Student-t is more robust. The dashed Uniform boundaries mark an infinite penalty outside the allowed interval.

Interactive: likelihood → loss

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/likelihood-to-loss

`likelihood-to-loss.html` — three synchronized panels: the **noise density** $p(r)$, the **loss** $-\log p(r)$, and **data with a fitted line**.
Controls: Gaussian / Laplace / Uniform / Student-t · noise scale · slope & intercept · **add an outlier** · per-point contributions.

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Ask: which model reacts most to an outlier? why does Uniform give an infinite barrier? which loss is robust?

Classification: cross-entropy

Classification should output probabilities

For K classes:

$$p_{\boldsymbol{\theta}}(Y = k \mid \boldsymbol{x}) \geq 0, \qquad \sum_{k=1}^{K} p_{\boldsymbol{\theta}}(Y = k \mid \boldsymbol{x}) = 1$$

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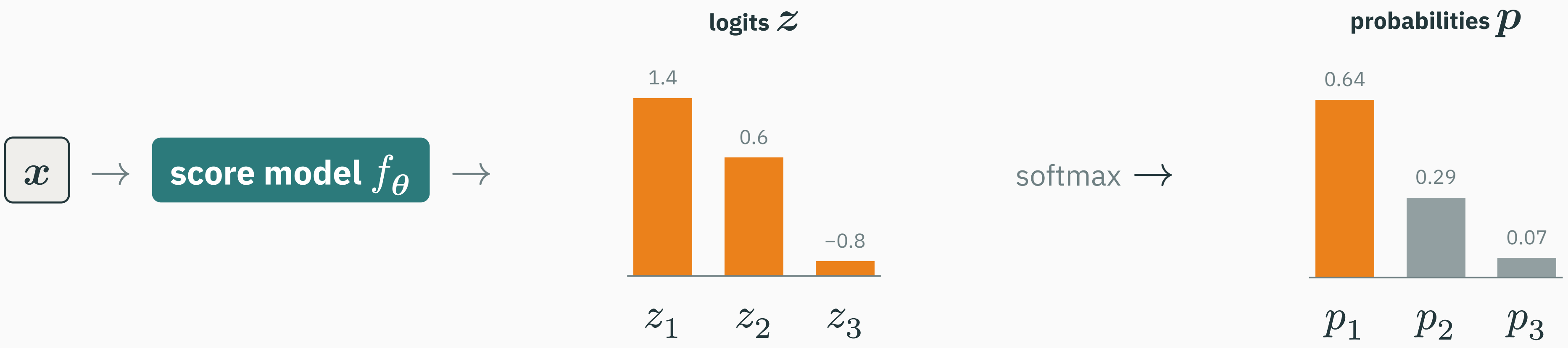

$f_{\boldsymbol{\theta}}$ may be **linear** (softmax regression) or **nonlinear** (a neural network).
It produces **logits** $z_k \in \mathbb{R}$; softmax converts them to **probabilities** $p_k \in [0, 1]$.

Binary classification: Bernoulli model

$$Y \mid \boldsymbol{X} = \boldsymbol{x} \sim \mathrm{Bernoulli}(p_{\boldsymbol{\theta}}(\boldsymbol{x}))$$

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$p_{\boldsymbol{\theta}}(y \mid \boldsymbol{x}) = p^y(1-p)^{1-y}$

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$$y = 1 \Rightarrow p(y) = p$$

$$y = 0 \Rightarrow p(y) = 1 - p$$

Bernoulli NLL gives BCE

D DERIVATION

$$-\log p_{\boldsymbol{\theta}}(y \mid \boldsymbol{x}) = -\log\left(p^{y}(1-p)^{1-y}\right)$$

Bernoulli NLL gives BCE

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$$-\log p_{\boldsymbol{\theta}}(y \mid \boldsymbol{x}) = -\log\left(p^y (1-p)^{1-y}\right)$$

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Binary cross-entropy = Bernoulli NLL.

Check. True label $y = 1$: if $p = 0.8$ the loss is $-\log 0.8 \approx 0.22$; if $p = 0.2$ it is $-\log 0.2 \approx 1.61$.

Why use logits?

V VISUAL

The network emits an unbounded score $z = f_{\boldsymbol{\theta}}(\boldsymbol{x}) \in \mathbb{R}$; the **sigmoid** maps it to a probability:

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}$$

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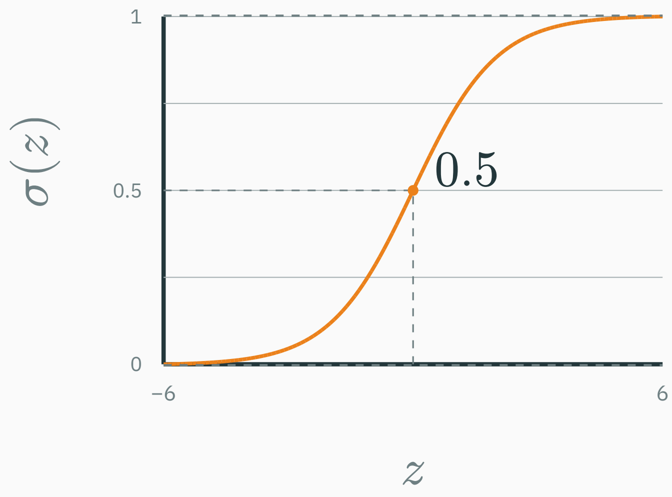
$$p = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$z \to -\infty : p \to 0$$
$$z = 0 : p = 0.5$$
$$z \to +\infty : p \to 1$$

In practice sigmoid + BCE are fused for numerical stability.



Logistic regression = Bernoulli MLE

D DERIVATION

Use an augmented input $\tilde{\boldsymbol{x}}_i$ so $\boldsymbol{\theta}$ contains both weights and the intercept:

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Logistic regression is Bernoulli maximum likelihood; its training objective is **binary cross-entropy**.

Multi-class model and softmax

V VISUAL

CATEGORICAL OBSERVATION

$$Y \mid \boldsymbol{X} = \boldsymbol{x} \sim \text{Categorical}(p_1, \ldots, p_K)$$

A **one-hot** target has one 1 at the true class and 0 everywhere else.

Example: class 2 of 3 $\longrightarrow \boldsymbol{y} = (0, 1, 0)$

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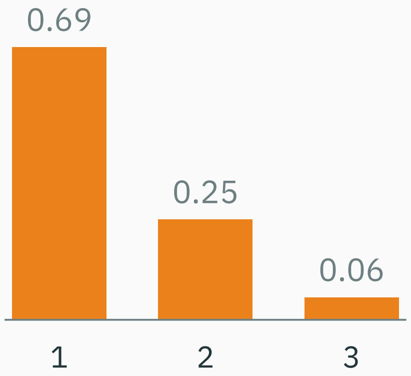
NETWORK PARAMETERIZATION

$$\boldsymbol{z} = f_{\boldsymbol{\theta}}(\boldsymbol{x}), \quad p_k = \frac{e^{z_k}}{\sum_j e^{z_j}}$$

softmax probabilities



Multi-class model and softmax

V VISUAL

CATEGORICAL OBSERVATION

$$Y \mid \boldsymbol{X} = \boldsymbol{x} \sim \mathrm{Categorical}(p_1, \ldots, p_K)$$

A **one-hot** target has one 1 at the true class and 0 everywhere else.

Example: class 2 of 3 $\longrightarrow \boldsymbol{y} = (0, 1, 0)$

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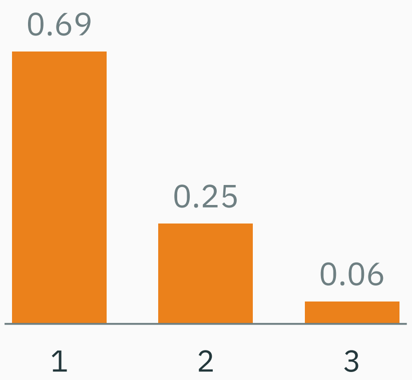

Softmax guarantees $p_k > 0$ and $\sum_k p_k = 1$, as a categorical model requires.

Categorical NLL gives cross-entropy

D DERIVATION

$$-\log p_{\boldsymbol{\theta}}(\boldsymbol{y} \mid \boldsymbol{x}) = -\log \prod_k p_k^{y_k} = -\sum_k y_k \log p_k$$

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D DERIVATION

$$-\log p_{\boldsymbol{\theta}}(\boldsymbol{y} \mid \boldsymbol{x}) = -\log \prod_k p_k^{y_k} = -\sum_k y_k \log p_k$$

Because $\boldsymbol{y}$ is one-hot, only the true class survives:

$$\mathcal{L} = -\log p_{\text{true class}}$$

Confident mistakes are expensive

V VISUAL

$$\ell(p_y) = -\log p_y$$

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Read off the loss at three confidences for the **true** class:

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VISUAL

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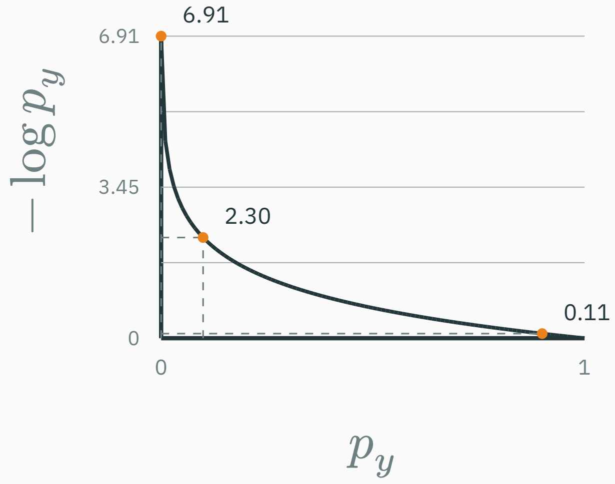
Read off the loss at three confidences for the **true** class:

$$p_y = 0.9 \Rightarrow \ell \approx 0.105$$
$$p_y = 0.1 \Rightarrow \ell \approx 2.303$$
$$p_y = 0.001 \Rightarrow \ell \approx 6.908$$



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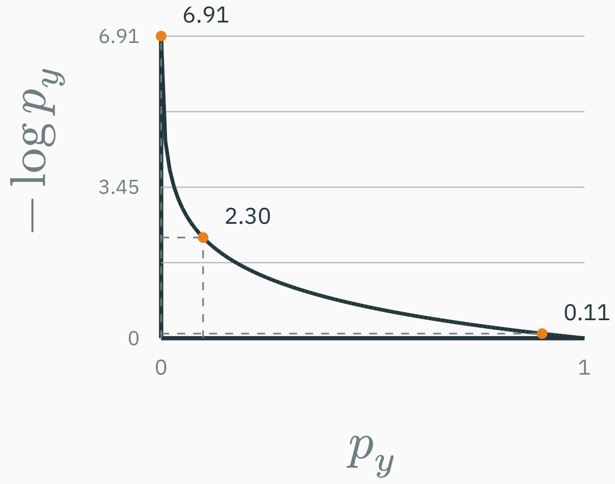

The negative log makes a confident wrong prediction a large loss.

Checkpoint: confident mistakes

Q QUESTION

Why does cross-entropy penalize a confidently wrong prediction strongly?

A. It makes all class probabilities equal

B. The true-class probability is near zero, so $-\log p_y$ is large

C. It adds an L_2 penalty to the logits

D. The gradient is always zero

Answer: confident mistakes

A ANSWER

Correct: B — The true-class probability is near zero

Why: The negative log grows without bound as p_y approaches zero, matching the fact that the model declared the observed class nearly impossible.

A binary label takes only two values

★ OPTIONAL

For one observed example, let the target be $y = 1$ and the model predict

$$p = P(Y = 1|\boldsymbol{x}) = 0.8.$$

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Squared error

$$\ell_{\text{MSE}} = (y - p)^2 = (1 - 0.8)^2 = 0.04.$$

Its usual likelihood interpretation is Gaussian:

$$Y|\boldsymbol{x} \sim \mathcal{N}(0.8, \sigma^2).$$

Permits all real Y—not just labels.

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Binary cross-entropy

$$\ell_{\text{CE}} = -\log p = -\log 0.8 \approx 0.223.$$

Its likelihood is Bernoulli:

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MSE is possible; cross-entropy is the negative log-likelihood matched to a binary observation.

Bayes’ rule for learning

Bayes' rule: from events to model parameters

V VISUAL

PROBABILITY LECTURE

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

A — event or hypothesis to infer

B — observed evidence

Examples: disease given a test; urn given a sampled colour.

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MACHINE / DEEP LEARNING

$$p(\boldsymbol{\theta} \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{D})}$$

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$\mathcal{D}$ — observed training dataset

Question: which parameter settings remain plausible after seeing the data?

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Question: which parameter settings remain plausible after seeing the data?

Replace A by θ and B by $\mathcal{D}$: the algebra is unchanged; the interpretation changes.

Three heads do not prove $\theta = 1$

Q QUESTION

Let $\theta = P(H)$ and suppose the first three flips are $\mathcal{D} = (H, H, H)$.

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H · H · H

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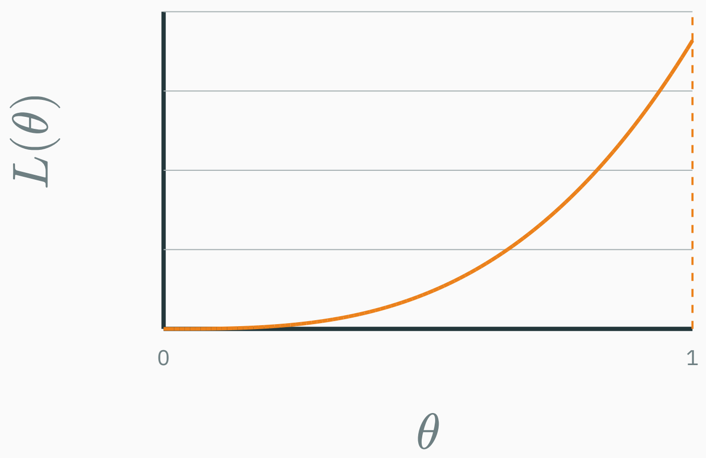
$$L(\theta) = \theta^3$$



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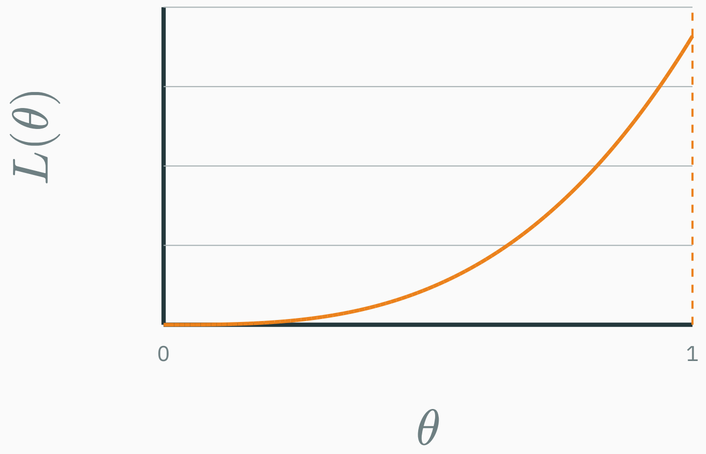

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$$\hat{\theta}_{\text{MLE}} = \arg\max_{0 \le \theta \le 1} \theta^3 = 1$$

Did we learn that the coin **always** lands heads—or did a merely head-biased coin produce a lucky run?

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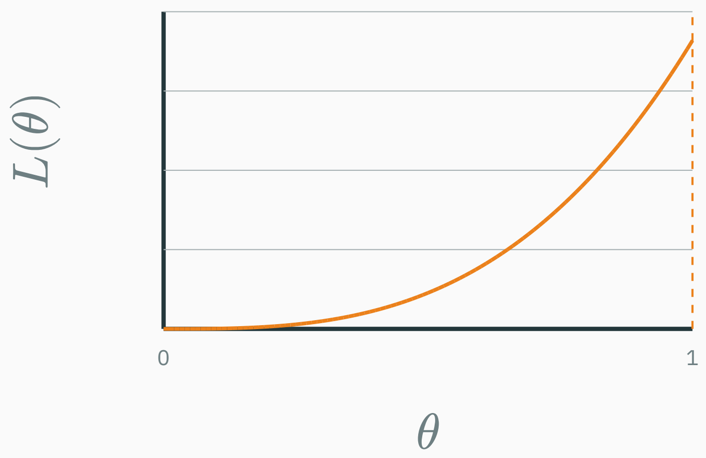

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MLE identifies the best-fitting θ; it does not make $\theta = 1$ certain.

A prior tempers the conclusion from three flips

V VISUAL

Choose a smooth prior on $\theta \in [0, 1]$, highest near the fair-coin value 0.5.

Prior

$p(\theta) = 6\theta(1 - \theta)$

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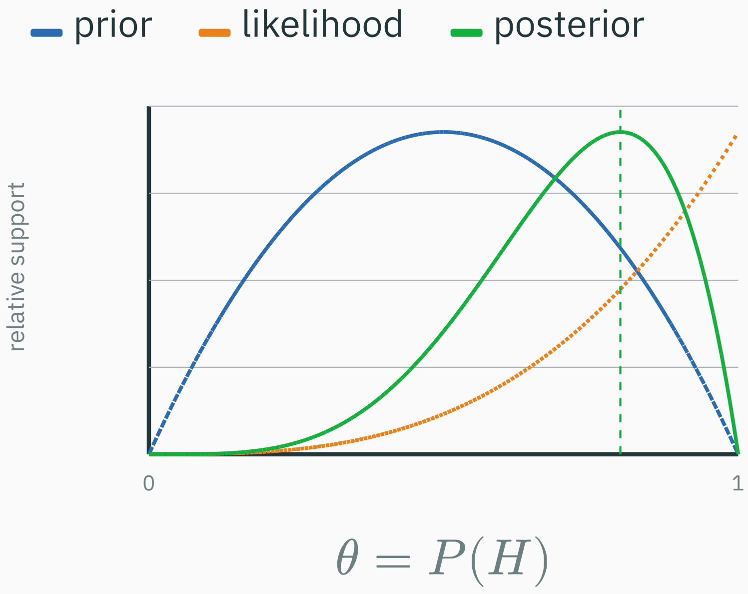
Posterior

$$p(\theta \mid \mathcal{D}) \propto \theta^3 p(\theta) \propto \theta^4(1-\theta)$$

Likelihood peak (MLE): $\theta = 1$

Posterior peak: $\theta = \frac{4}{5} = 0.8$



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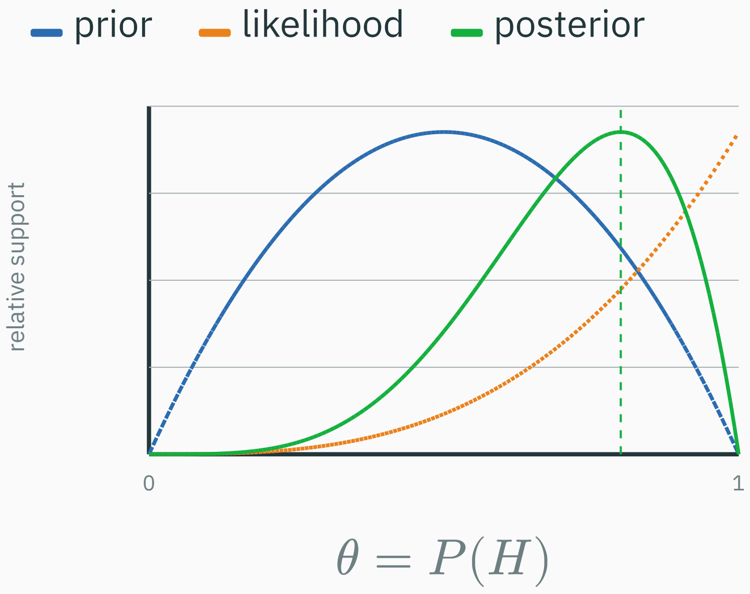

With little data, the prior matters; as observations accumulate, the likelihood increasingly dominates.

Priors, MAP and regularization

Every learner has an inductive bias

V VISUAL

When several predictors fit the training data, the learner still needs a preference for **unseen inputs**. That preference is its **inductive bias**.

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Model	Built-in preference $\to$ consequence
Linear regression	responses vary linearly with features $\to$ line or plane extrapolation
k**-NN**	nearby inputs tend to have similar outputs $\to$ local prediction
Decision tree	a few axis-aligned rules partition the space $\to$ piecewise-constant regions
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A prior is one explicit inductive bias; the model class, features, architecture, and optimization add others.

The same evidence can start from different priors

D DERIVATION

Suppose only two coins are possible: **fair** $F : \theta = 0.5$ or **heads-biased** $B : \theta = 0.9$.

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Observe $\mathcal{D} = (H, H, H)$: $P(\mathcal{D}|F) = 0.5^3 = 0.125 \quad P(\mathcal{D}|B) = 0.9^3 = 0.729.$

Prior A: probably fair

$P(F) = 0.90, \quad P(B) = 0.10$

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Prior B: probably biased

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prior × likelihood $F: \quad 0.10 \times 0.125 = 0.0125$

$B: \quad 0.90 \times 0.729 = 0.6561$

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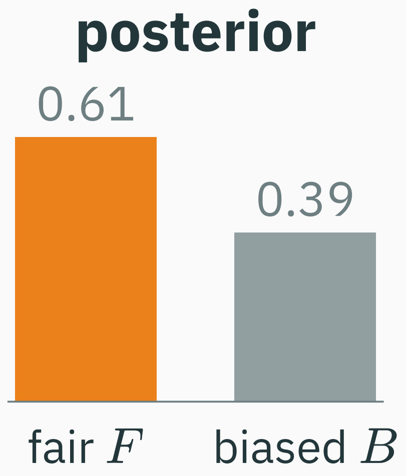
Bayes' rule first forms a weight for each hypothesis: prior × likelihood.

Normalize the weights to obtain the posterior

VISUAL

Divide each weight by the sum of the two weights so the posterior probabilities add to 1.

Prior A: probably fair



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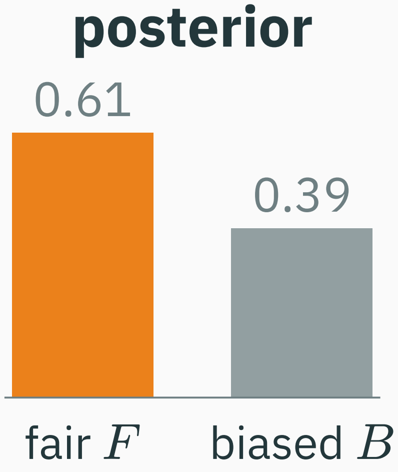
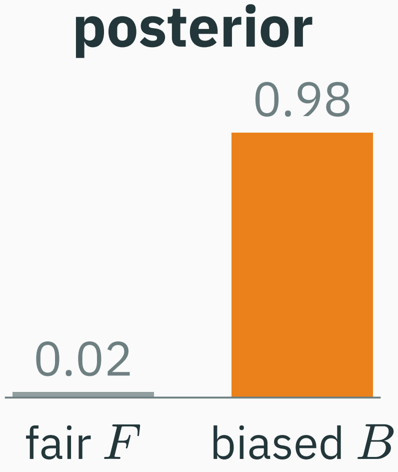


Prior B: probably biased



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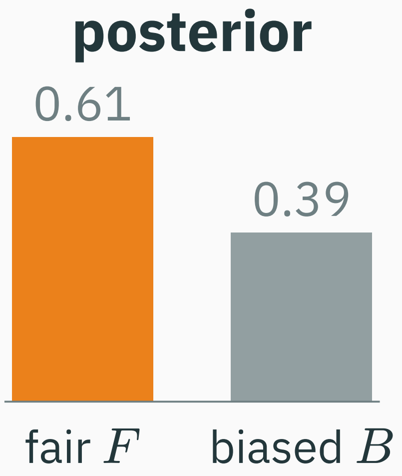

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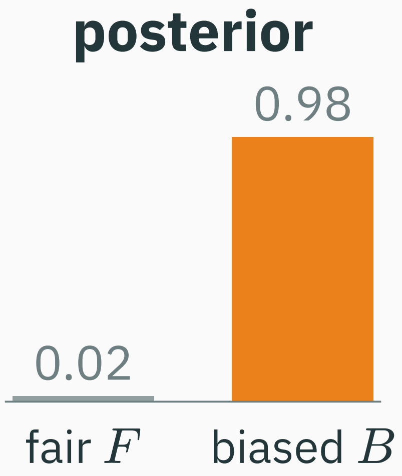


The observation is identical; with little data, different priors produce different posteriors.

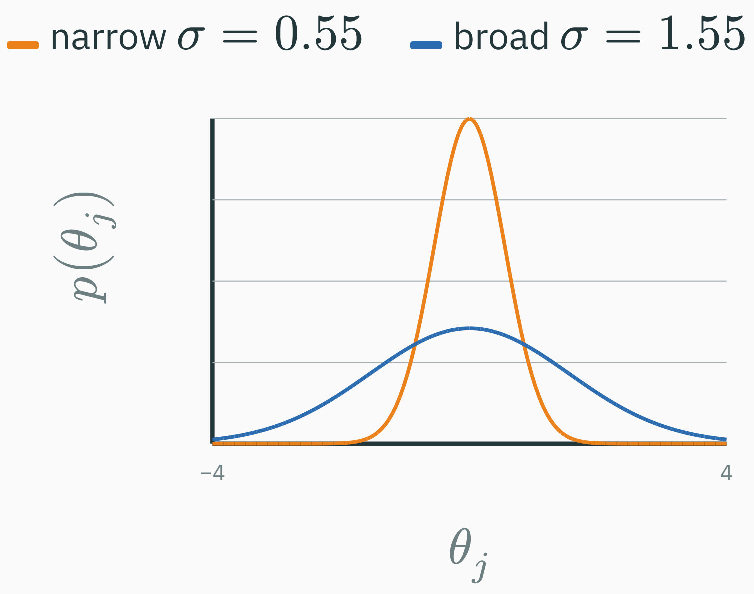
A neural-network prior ranks plausible parameter vectors

V VISUAL

$\boldsymbol{\theta} \sim p(\boldsymbol{\theta})$

For a neural network, $\boldsymbol{\theta}$ contains all weights and biases. Before using the current dataset, a prior states which settings are more plausible.

It can prefer small or sparse weights, smooth functions, and correlated parameters.



narrow = stronger preference near 0

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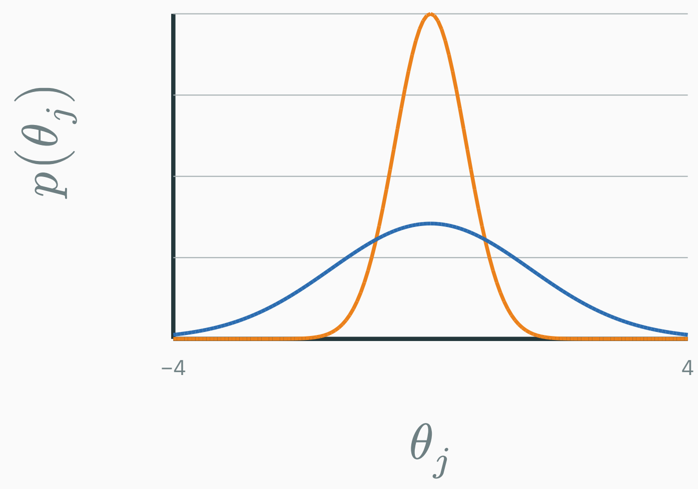

narrow = stronger preference near 0

A prior makes part of the model's inductive bias explicit as a probability distribution.

Parameter priors have visible consequences

V VISUAL

REGRESSION

$$Y_i \mid \boldsymbol{X}_i = \boldsymbol{x}_i \sim \mathcal{N}(\boldsymbol{\theta}^\top \tilde{\boldsymbol{x}}_i, \sigma^2)$$

$$\boxed{p(\boldsymbol{\theta})} \longrightarrow \boxed{\text{small } \|\boldsymbol{\theta}\|} \longrightarrow \boxed{\text{smaller slopes}}$$

Before seeing data, flatter linear predictions are more plausible.

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BINARY CLASSIFICATION

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Before seeing data, extremely confident predictions are less plausible.

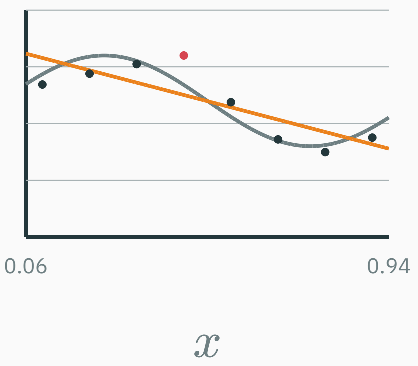
The prior acts on parameters; its visible effect depends on how those parameters enter the model.

Model complexity: underfitting to overfitting

Fit the same eight observations by least squares; only the polynomial degree changes.

dashed = underlying trend · orange = fitted polynomial · red dot = one noisy observation

degree 1



underfit

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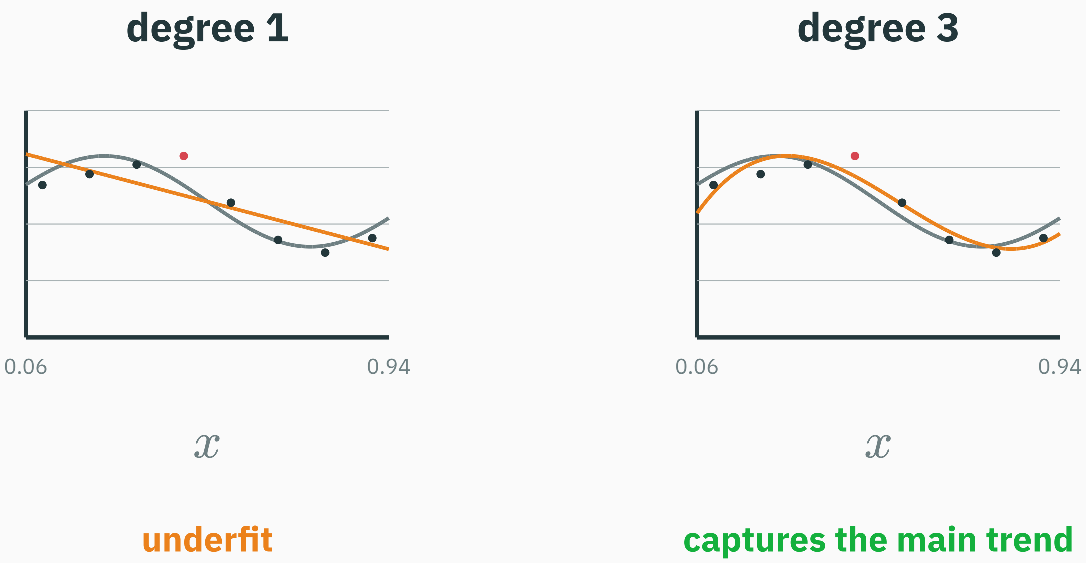

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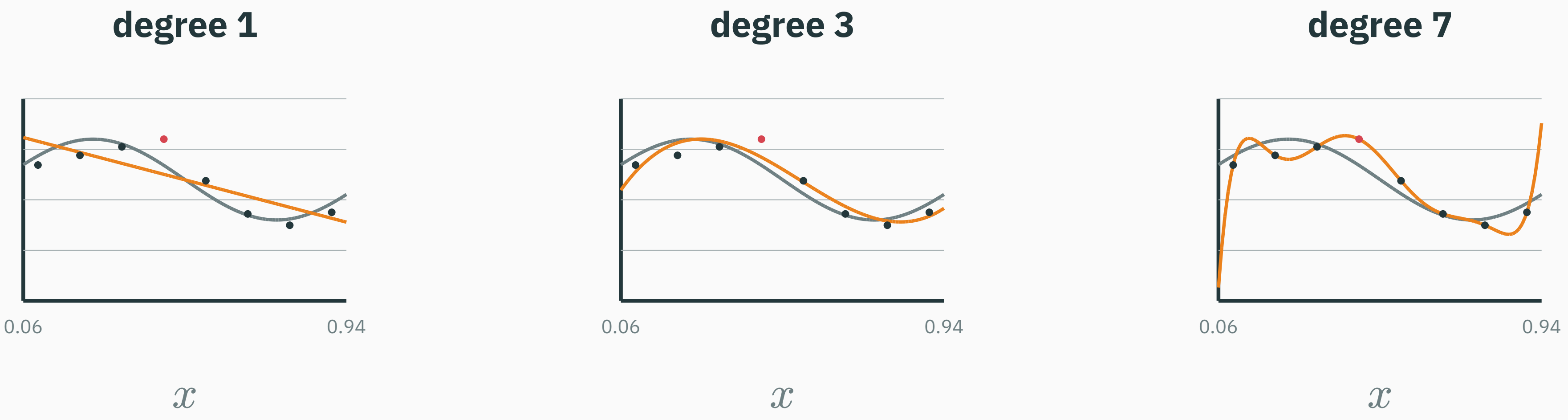

underfit

captures the main trend

overfit

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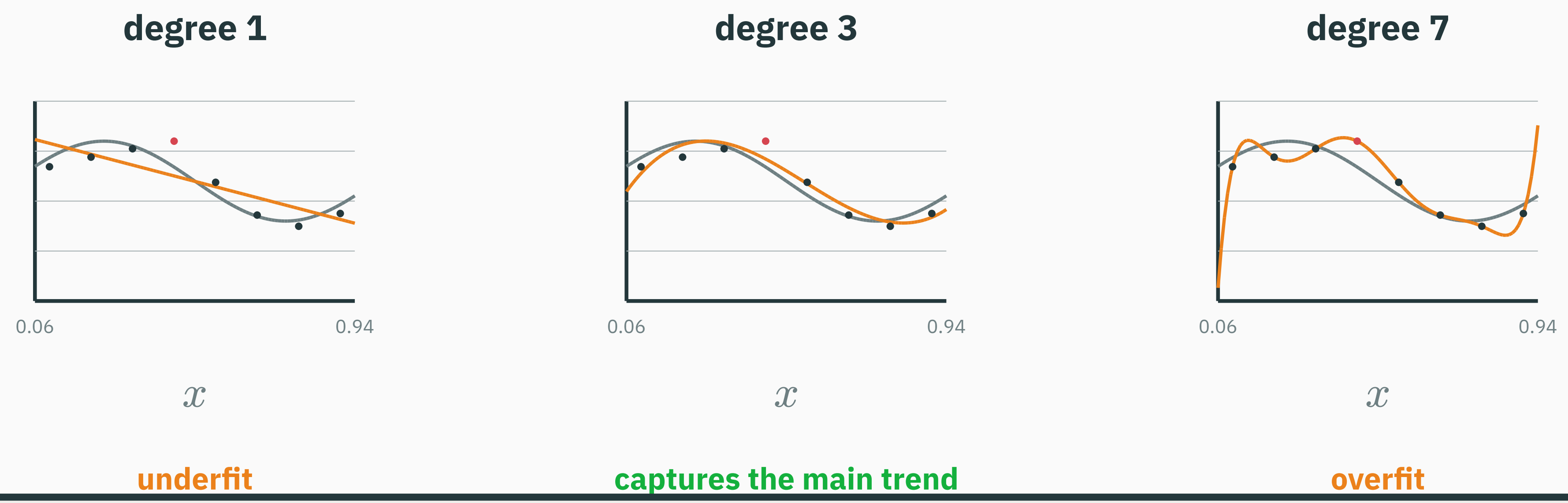

Higher degree lowers training error, but degree 7 chases the noisy observation and becomes unstable. Priors and regularization can prefer smoother solutions.

Start with four noisy points from a straight line

V VISUAL

The data were generated from $f_{\text{true}(x)} = 2 + 3x$ with small measurement noise.

x	$f_{\text{true}(x)}$	**observed** y
-1	-1	-0.8
0	2	1.8
1	5	5.3
2	8	7.9

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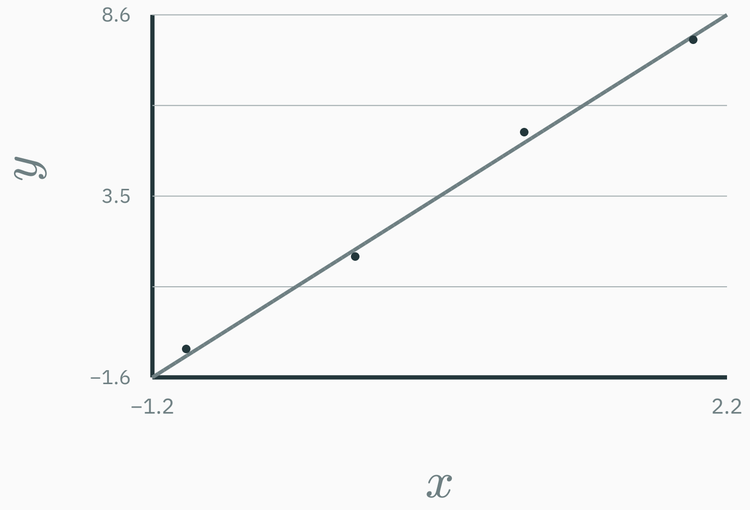

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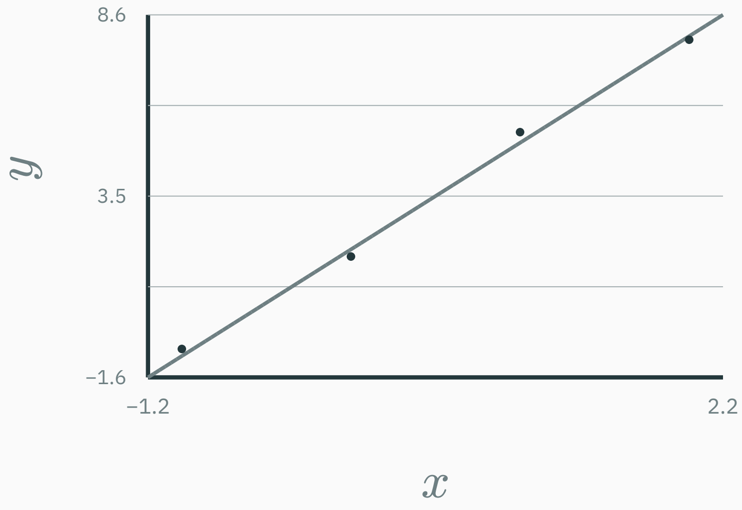

grey line = generating rule · dots = observed data

The learner sees only the four observations and must infer the intercept b and slope w.

The likelihood peaks near the generating line

D DERIVATION

Model each observation as

$$Y_i | x_i, b, w \sim \mathcal{N}(b + wx_i, 0.5^2).$$

The data score each candidate line

$$L(b, w) \propto \exp\left(-\frac{1}{2(0.5)^2} \sum_i (y_i - b - wx_i)^2\right).$$

candidate (b, w)	squared error
$(2, 3)$	0.180
$(0, 1)$	56.580
MLE $(2.07, 2.96)$	0.162

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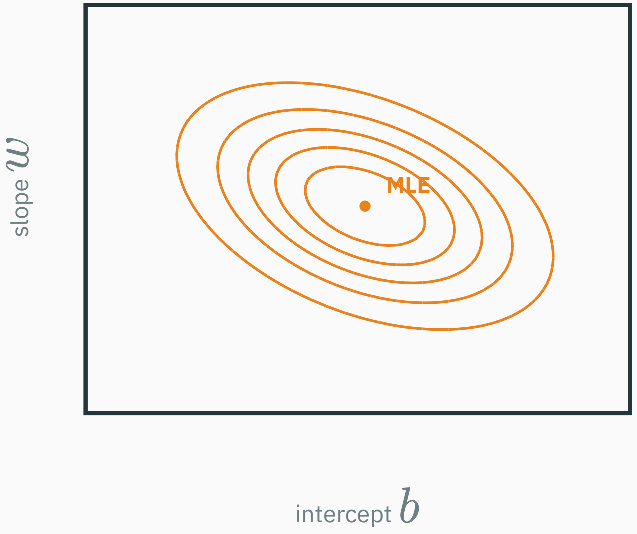

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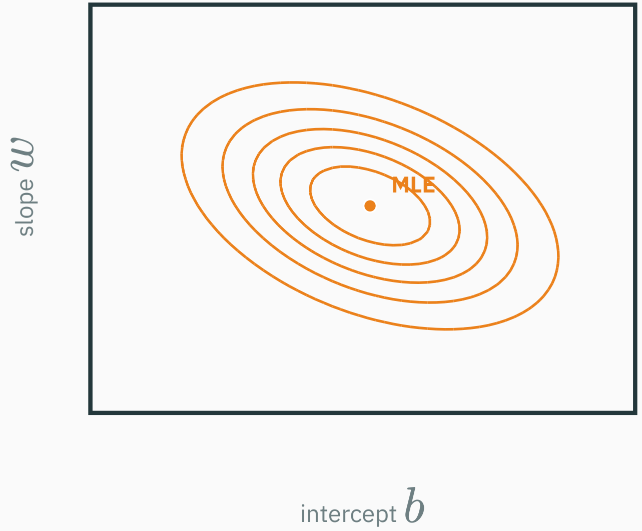

The observed data make intercepts near 2 and slopes near 3 most plausible.

Before data, the prior prefers parameters near zero

V VISUAL

Choose a simple Gaussian prior over the two regression parameters:

$$\boldsymbol{\theta} = (b, w) \sim \mathcal{N}\big(\mathbf{0}, 0.5^2 \boldsymbol{I}\big).$$

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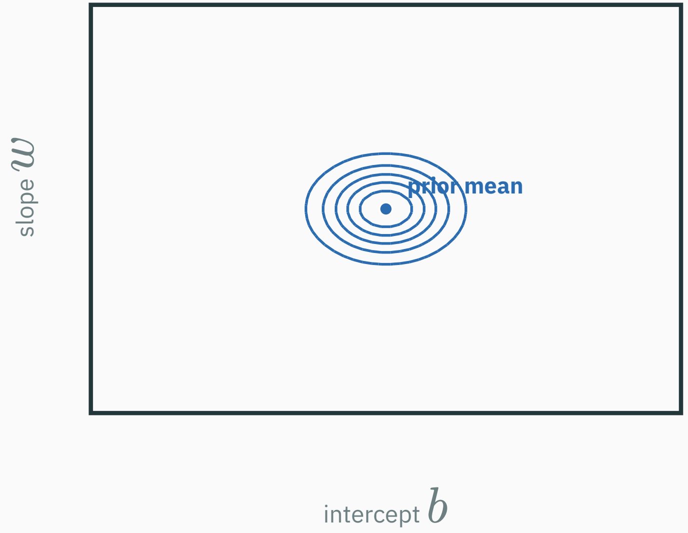

What does it express?

$-\log p(b, w) = \frac{1}{2(0.5)^2}(b^2 + w^2) + C = 2(b^2 + w^2) + C.$

Before seeing this dataset:

- $b \approx 0$: intercept near zero
- $w \approx 0$: a flatter line
- farther from $(0, 0)$: less prior density

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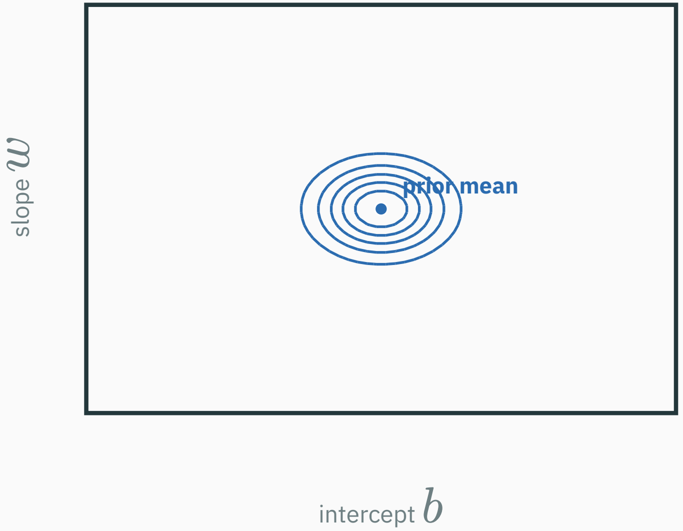

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The likelihood says “fit these data”; the prior says “prefer smaller coefficients”.

Bayes combines the regression likelihood and prior

$$p(b, w \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid b, w)\, p(b, w)}{p(\mathcal{D})}$$

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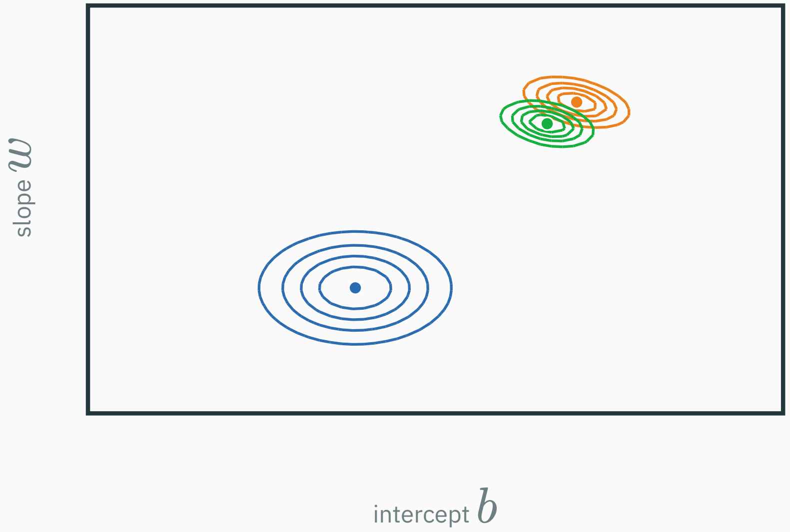


Contours: prior likelihood posterior

● MLE $(2.07, 2.96)$ ● posterior peak $(1.79, 2.62)$ ● prior mean $(0, 0)$

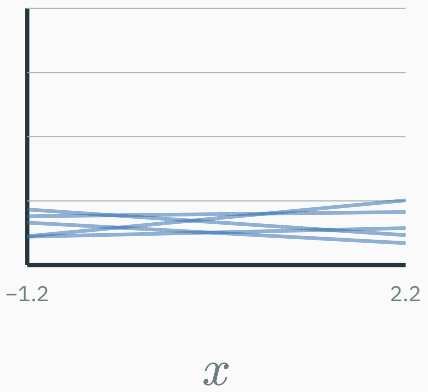
The narrower prior visibly pulls the posterior peak away from the MLE and toward zero.

The prior changes the fitted line

V VISUAL

Prior before data

possible lines; not fitted



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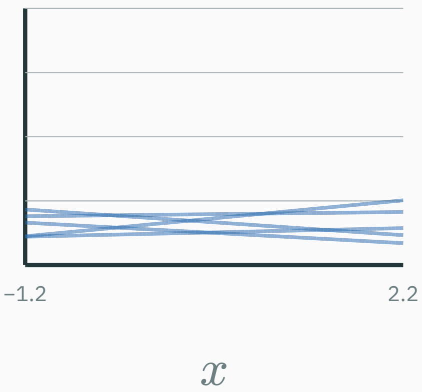
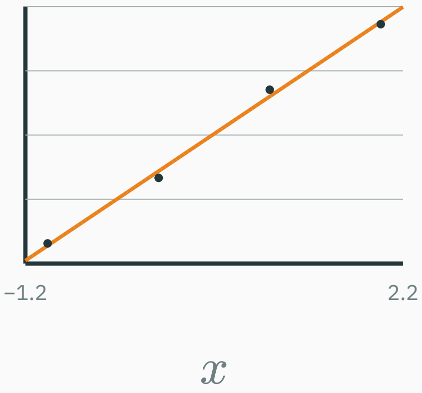


MLE

$$\hat{y} = 2.07 + 2.96x$$



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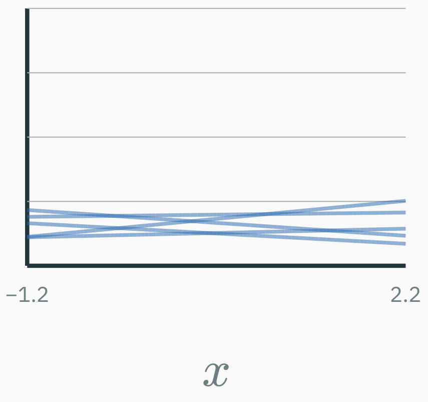

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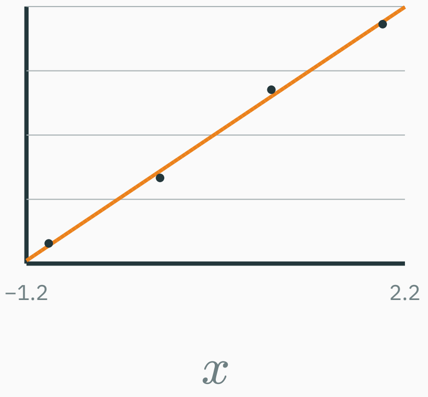
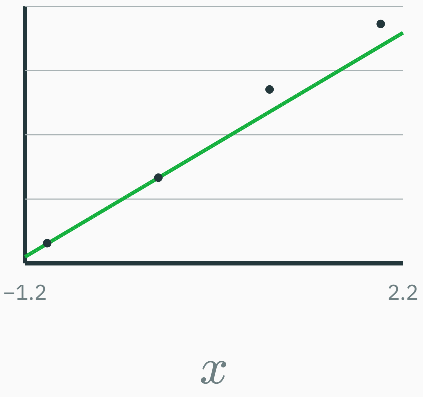


MAP

$$\hat{y} = 1.79 + 2.62x$$



The prior changes the fitted line

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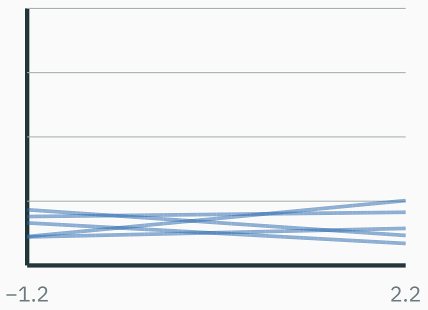

x

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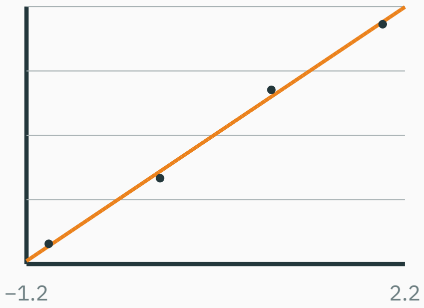

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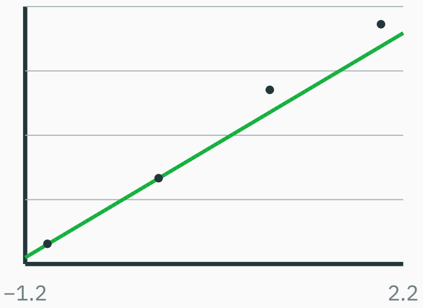

x

MLE follows these observations most closely; MAP accepts a little more error to honour the prior preference for smaller coefficients.

MAP estimation

D DERIVATION

$$\hat{\boldsymbol{\theta}}_{\mathrm{MAP}} = \arg\max_{\boldsymbol{\theta}} p(\boldsymbol{\theta} \mid \mathcal{D}) = \arg\max_{\boldsymbol{\theta}} p(\mathcal{D} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})$$

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Take negative logs:

$$\hat{\boldsymbol{\theta}}_{\mathrm{MAP}} = \arg\min_{\boldsymbol{\theta}} \left(\underbrace{-\log p(\mathcal{D} \mid \boldsymbol{\theta})}_{\text{NLL}} + \underbrace{-\log p(\boldsymbol{\theta})}_{\text{negative log-prior}} \right)$$

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MAP objective: NLL + negative log-prior

In ML language: data loss + regularization.

A Gaussian prior is L_2 regularization

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Assume the parameters are centred at zero before seeing the data:

$$\boldsymbol{\theta} \sim \mathcal{N}(\mathbf{0}, \tau^2 \boldsymbol{I}).$$

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$$\Rightarrow \quad \mathcal{L}_{\text{MAP}} = \text{NLL} + \underbrace{\frac{1}{2\tau^2}}_{\lambda}\|\boldsymbol{\theta}\|_2^2 + C$$

$\lambda = \frac{1}{2\tau^2}$: smaller $\tau \to$ larger $\lambda \to$ stronger shrinkage toward 0.

Prior width controls how far MAP moves

V VISUAL

Same likelihood: $L(\theta) \propto \mathcal{N}\left(\theta; \hat{\theta}_{\text{MLE}} = 2, s^2\right)$ with $s = 0.6$; prior: $p(\theta) = \mathcal{N}(0, \tau^2)$.

$\hat{\theta}_{\text{MAP}} = \frac{\tau^2}{\tau^2+s^2}\hat{\theta}_{\text{MLE}}$ (a shrinkage factor between 0 and 1)

prior — likelihood — posterior

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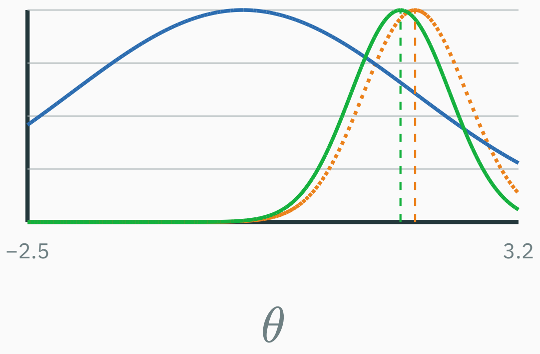
prior likelihood posterior

Broad prior: $\tau = 2, \lambda = 0.125$

MLE $= 2 \to \text{MAP} \approx 1.83$



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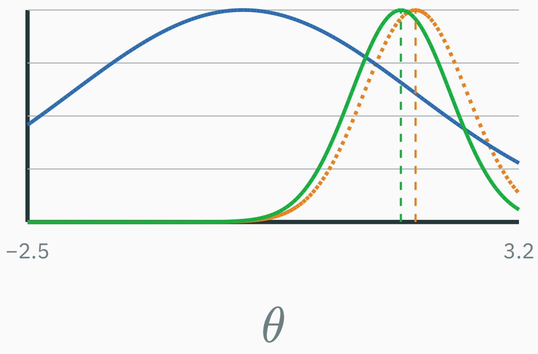
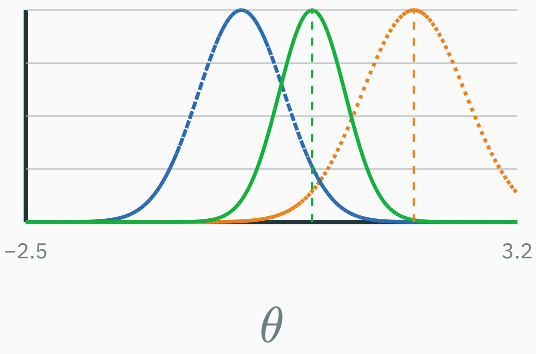


Narrow prior: $\tau = 0.5, \lambda = 2$

MLE $= 2 \rightarrow \mathrm{MAP} \approx 0.82$



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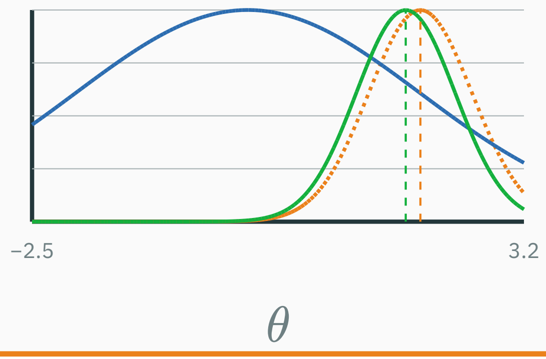

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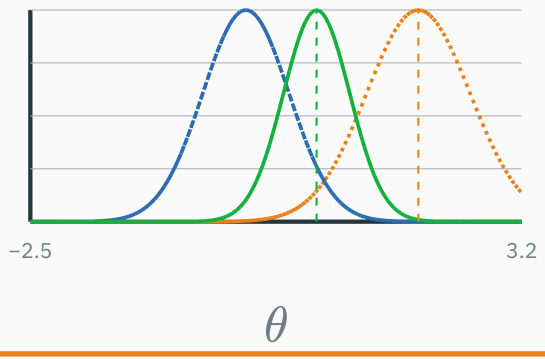

The narrower prior gives the larger λ and pulls MAP farther from the MLE toward zero.

A multivariate Gaussian prior has geometry

V VISUAL

$$\boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \qquad -\log p(\boldsymbol{\theta}) = \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\theta} - \boldsymbol{\mu}) + C$$

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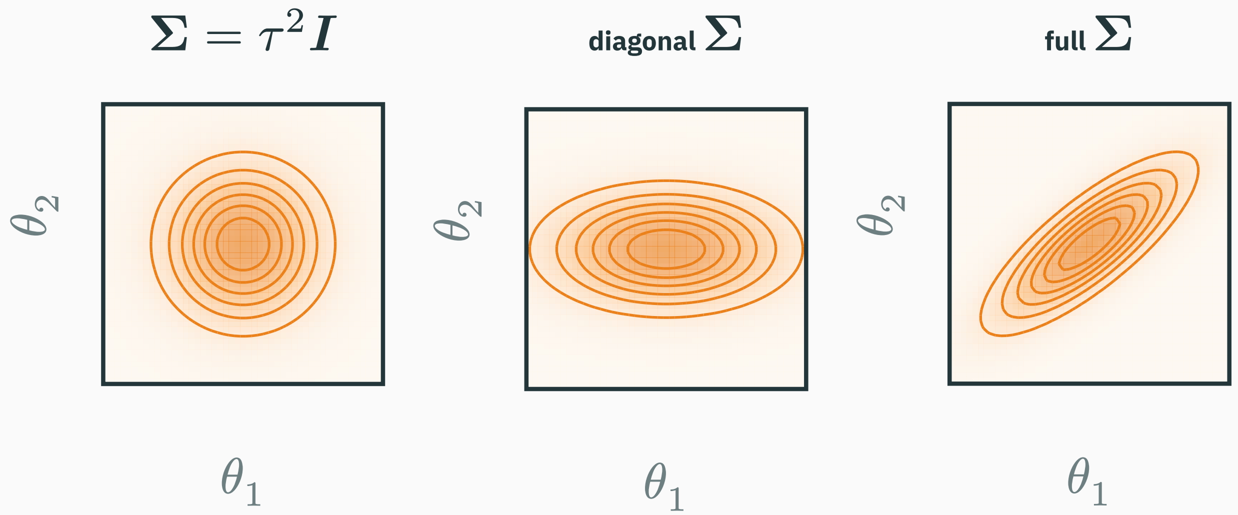

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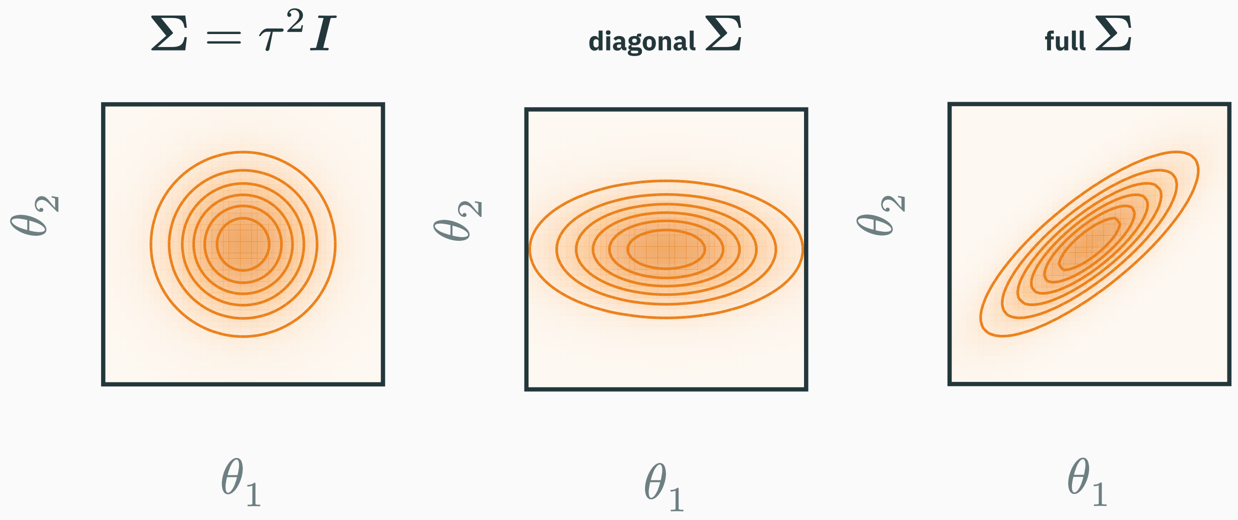

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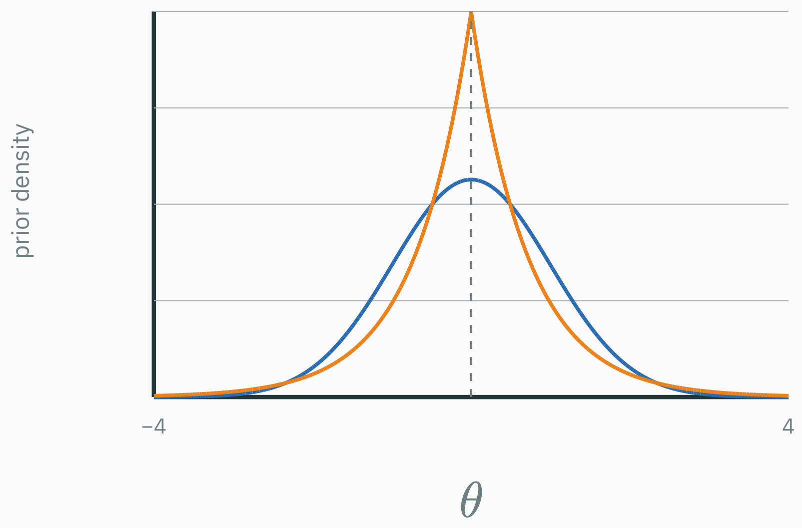
Covariance says **which directions in parameter space are plausible.**

Laplace is sharper at zero than a Normal prior

V VISUAL

Compare zero-centred priors with the **same variance**, 1:

Normal: $\sigma = 1$ Laplace: $b = \frac{1}{\sqrt{2}}$



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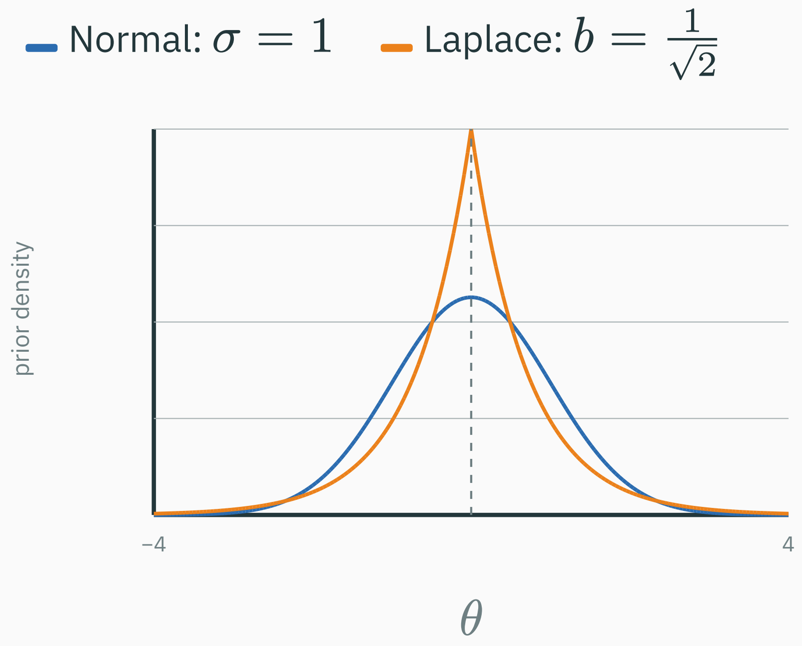

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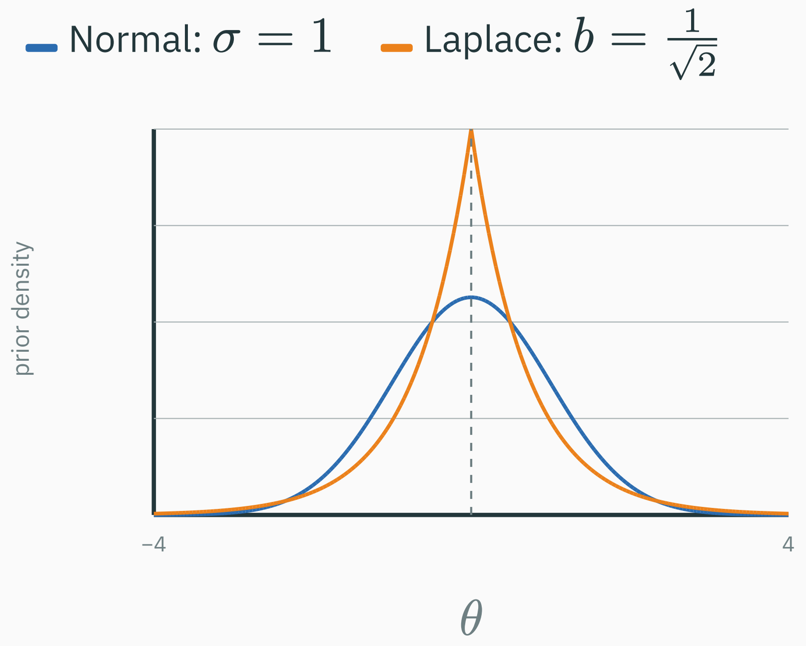

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The Laplace peak creates a corner in the negative log-prior—the source of L_1 sparsity.

The L_1 corner can set a coefficient exactly to zero

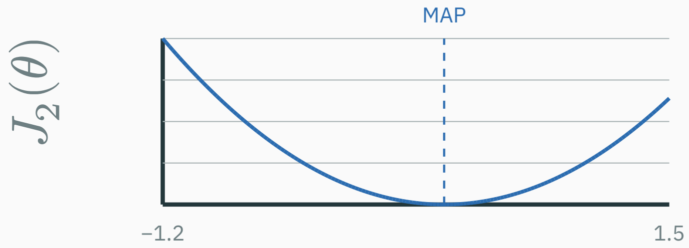
D DERIVATION

For one coefficient, let the data term be $D(\theta) = \frac{1}{2}(\theta - 0.6)^2$, centred at $z = 0.6$.

Normal prior $\to L_2$

$J_2(\theta) = D(\theta) + \frac{1}{2}\theta^2.$

$$\hat{\theta}_{\text{MAP}} = \frac{0.6}{1+1} = 0.3.$$



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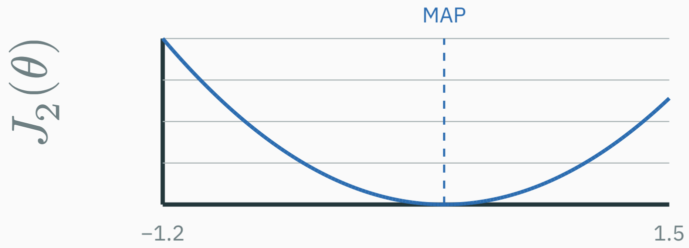
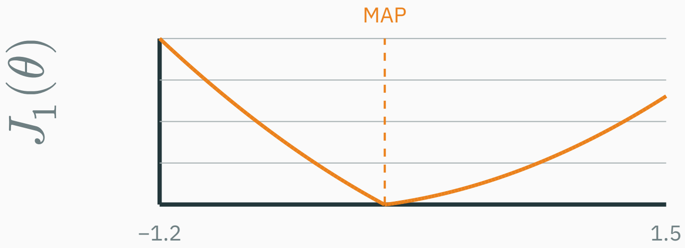


Laplace prior $\to L_1$

$$J_1(\theta) = D(\theta) + |\theta|.$$

$$\hat{\theta}_{\text{MAP}} = \operatorname{sign}(z)\max(|z| - 1, 0) = 0.$$



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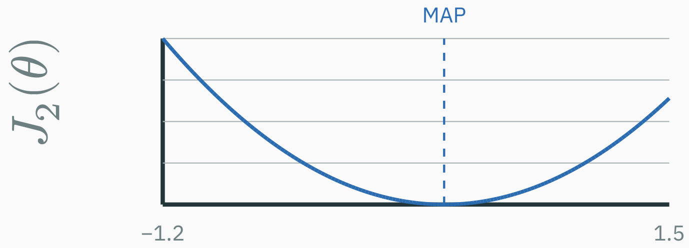

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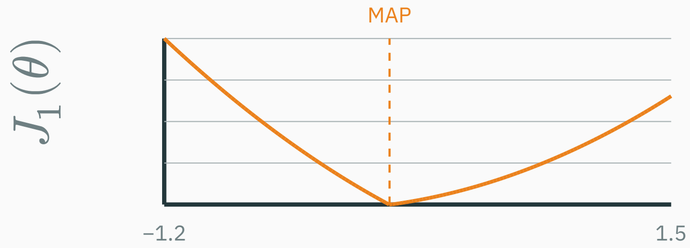

Applied coefficient by coefficient, L_1 can make many MAP estimates exactly zero: sparsity.

Prior strength controls underfitting and overfitting

V VISUAL

For the summed-NLL convention used above, $\lambda = \frac{1}{2\tau^2}$:

weak prior $\lambda \approx 0$ → balanced → strong prior large λ

Prior strength controls underfitting and overfitting

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$$\boxed{\begin{array}{c}\text{weak prior}\\ \lambda \approx 0\end{array}} \longrightarrow \boxed{\text{balanced}} \longrightarrow \boxed{\begin{array}{c}\text{strong prior}\\ \text{large } \lambda\end{array}}$$

DATA FIT DOMINATES

flexible model
high variance · overfit

LIKELIHOOD + PRIOR

enough flexibility
better generalization

PRIOR DOMINATES

weights forced small
high bias · underfit

Prior strength controls underfitting and overfitting

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DATA FIT DOMINATES	LIKELIHOOD + PRIOR	PRIOR DOMINATES
flexible model	enough flexibility	weights forced small
high variance · overfit	**better generalization**	**high bias · underfit**

Regularization trades variance for bias; more is not always better.

Interactive: likelihood, prior, MAP

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/map-regularization

map-regularization.html — a two-parameter linear model so contours are visible. Shows **likelihood**, **prior**, and **posterior** contours with the **MLE**, **MAP**, and zero points.
Controls: amount of data · noise · prior variance · Gaussian vs Laplace · prior correlation.

Interactive: likelihood, prior, MAP

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Ask: more data? narrower prior? why does MAP → MLE as data grows? a correlated prior?

Full Bayes and deep learning

MAP is still one parameter vector

MLE $\hat{\theta}_{\mathrm{MLE}}$ — one point

MAP $\hat{\theta}_{\mathrm{MAP}}$ — one point

MAP is still one parameter vector

MLE $\hat{\theta}_{\mathrm{MLE}}$ — one point

MAP $\hat{\theta}_{\mathrm{MAP}}$ — one point

Full Bayes keeps the whole posterior $p(\boldsymbol{\theta} \mid \mathcal{D})$ — a **distribution**, not a point.

Posterior predictive distribution

$$p(y^\star \mid \boldsymbol{x}^\star, \mathcal{D}) = \int p_{\boldsymbol{\theta}}(y^\star \mid \boldsymbol{x}^\star)\, p(\boldsymbol{\theta} \mid \mathcal{D})\, \mathrm{d}\boldsymbol{\theta}$$

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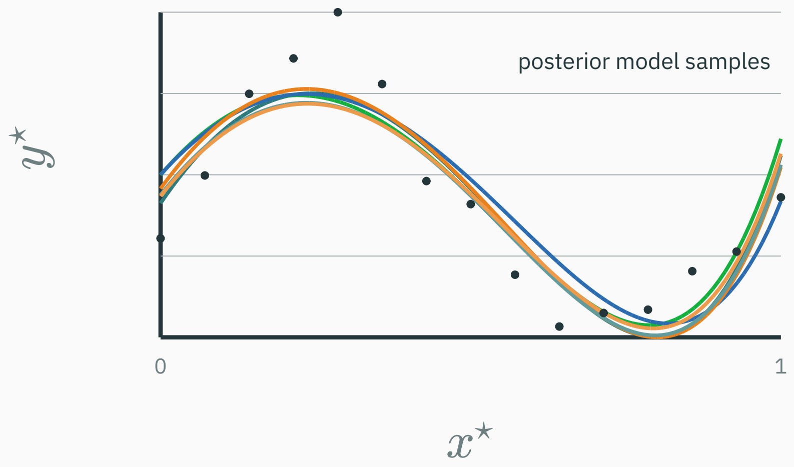

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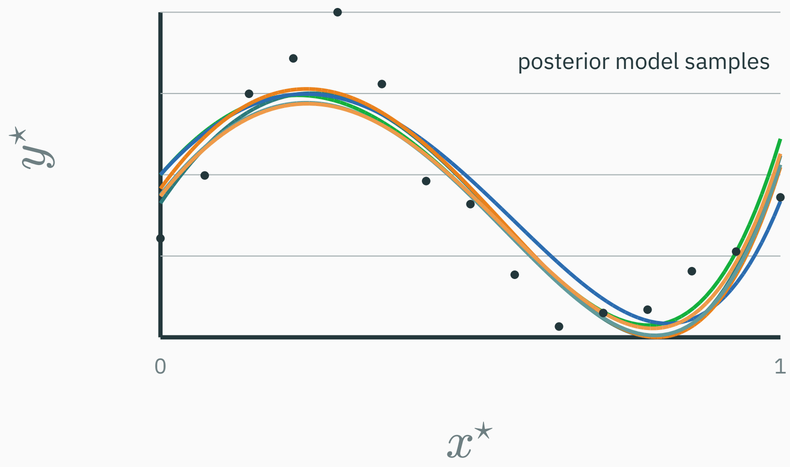

Bayesian prediction **averages over plausible models** — and reports uncertainty.

Why DL usually uses point estimates

★ OPTIONAL

A modern network has millions to billions of parameters, so integrating $p(\boldsymbol{\theta} \mid \mathcal{D})$ is hard.

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- deep **ensembles**
- **dropout**-based approximations
- **variational** inference

The standard supervised DL objective

$$\min_{\boldsymbol{\theta}} \underbrace{\frac{1}{|B|} \sum_{i \in B} -\log p_{\boldsymbol{\theta}}(y_i \mid \boldsymbol{x}_i)}_{\text{likelihood / data fit}} + \underbrace{\lambda \|\boldsymbol{\theta}\|_2^2}_{\text{prior / regularization}}$$

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DL doesn't discard probabilistic ML — it **scales** it with expressive functions and gradient descent.

Retrieval summary

V VISUAL

Gaussian likelihood $\Longrightarrow$ **MSE**
Categorical likelihood $\Longrightarrow$ **cross-entropy**
Gaussian prior $\Longrightarrow$ $\boldsymbol{L_2}$
Laplace prior $\Longrightarrow$ $\boldsymbol{L_1}$
MLE + prior $\Longrightarrow$ **MAP**

Every supervised objective begins with two assumptions

V VISUAL

MODEL

observation model

$$p_{\boldsymbol{\theta}}(y \mid \boldsymbol{x})$$

parameter prior

$$p(\boldsymbol{\theta})$$

$\rightarrow$

TRAIN

$$\mathcal{L}_{\mathrm{MAP}(\boldsymbol{\theta})} = \underbrace{\mathrm{NLL}(\mathcal{D};\boldsymbol{\theta})}_{\text{data term}} + \underbrace{\Omega(\boldsymbol{\theta})}_{\text{parameter term}}$$

$$\Omega(\boldsymbol{\theta}) = -\log p(\boldsymbol{\theta})$$

$$\hat{\boldsymbol{\theta}}_{\mathrm{MAP}} = \arg\min_{\boldsymbol{\theta}} \mathcal{L}_{\mathrm{MAP}}$$

$\rightarrow$

PREDICT

plug-in: fix $\theta = \hat{\theta}$

$$p_{\hat{\boldsymbol{\theta}}}(y^\star \mid \boldsymbol{x}^\star)$$

or average over the posterior

$$p(y^\star \mid \boldsymbol{x}^\star, \mathcal{D})$$

Loss $\leftrightarrow$ likelihood; regularizer $\leftrightarrow$ prior; prediction $\leftrightarrow$ plug-in or posterior predictive.

Next: from linear models to neural networks — **neurons, nonlinearities, and multilayer perceptrons.**